Trust-Aware Enterprise Credit Risk Prediction via Variational Autoencoder in Supply Chain Finance

XIAOBO, GUO, Institute of Information Science, Beijing Jiaotong University; Data Management Department, China Minsheng Bank, China
LU-AN, DONG, Data Management Department, China Minsheng Bank, China
YOURU, LI*, College of Computer Science, Beijing University of Technology, China
YANBO, WANG, Data Intelligence Division, Longying Zhida (Beijing) Technology, China
PENG, ZHANG, Cyberspace Institute of Advanced Technology, Guangzhou University, China
ZHENFENG ZHU*, Institute of Information Science, Beijing Jiaotong University, China

The growing complexity of transaction networks, dynamic enterprise interactions, and rising financial uncertainties in modern supply chains have made accurate credit risk assessment for small and medium-sized enterprises (SMEs) in supply chain finance (SCF) a critical priority. However, traditional credit prediction models struggle to capture meaningful latent risk patterns and dynamic risk propagation within supply chain networks, limiting their effectiveness, robustness, and interpretability in supporting trustworthy decision-making in SCF ecosystems. To this end, we propose **TrustVAE**, a novel end-to-end generative paradigm that jointly models iterative node aggregation, trust-aware message passing, and spatiotemporal dynamics for credit risk assessment in SCF. Specifically, TrustVAE leverages the *Iterative Latent Risk Aggregation (ILRA)* module to extract latent risk representations from raw enterprise data in an unsupervised manner through deep nonlinear transformations and iterative clustering, capturing inherent structural properties that extend beyond the static financial indicators used in conventional models. Meanwhile, the *Bounded Graph Attention Network (BGAN)* module integrates trust-aware bounded confidence into graph-based risk propagation, enhancing the fidelity of diffusion modeling by suppressing risk transmission across low-trust enterprise edges. Furthermore, TrustVAE introduces the *Temporal Graph Autoregressive Flow* (TGAF) module, which couples temporal dependencies with autoregressive flow modeling to characterize credit risk evolution across multi-scale temporal horizons (e.g., daily, weekly, and monthly), jointly capturing dynamic patterns in both temporal and graph-structured domains. Extensive experiments on real-world datasets demonstrate that TrustVAE outperforms existing competitors, achieving average absolute improvements of 1.59%, 3.17%, 1.93%, and 2.11% in ROC-AUC, AR, Accuracy, and F1 Score, respectively. The unified design of TrustVAE strengthens the practical foundation of information systems in addressing credit risk within data-intensive, networked environments, offering novel insights into generative modeling for credit risk prediction.

CCS Concepts: • **Information systems → Data mining**; • **Applied computing → Online banking**.

*Corresponding authors.

Authors' Contact Information: XIAOBO, GUO, xb_guo@bjtu.edu.cn, Institute of Information Science, Beijing Jiaotong University; Data Management Department, China Minsheng Bank, Beijing, China; LU-AN, DONG, Data Management Department, China Minsheng Bank, Beijing, China; YOURU, LI*, liyouru@bjut.edu.cn, College of Computer Science, Beijing University of Technology, Beijing, China; YANBO, WANG, Data Intelligence Division, Longying Zhida (Beijing) Technology, Beijing, China; PENG, ZHANG, Cyberspace Institute of Advanced Technology, Guangzhou University, Guangdong, China; Zhenfeng Zhu*, zhfzhu@bjtu.edu.cn, Institute of Information Science, Beijing Jiaotong University, Beijing, China.

ACM 1556-472X/2026/8-ART
https://doi.org/10.1145/3840289

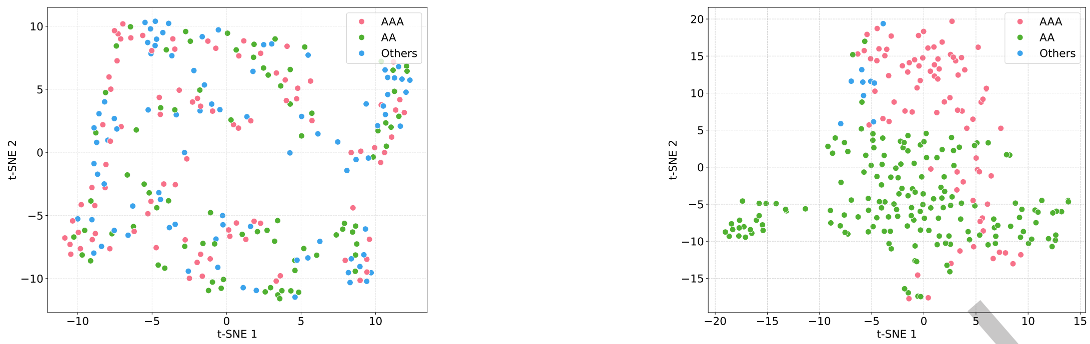
Additional Key Words and Phrases: iterative clustering, trust-aware bounded confidence, autoregressive flow, enterprise credit risk , supply chain finance, variational autoencoder.

1 Introduction

In recent years, with the continuous advancement of China's market economy reforms, competition among enterprises has intensified, making access to bank credit an essential factor for business growth, particularly for small and medium-sized enterprises (SMEs). As of 2024, SMEs constitute over 90% of all Chinese enterprises, contribute approximately 60% of GDP, and generate 80% of employment, underscoring their pivotal role in the national economy [40]. Despite their economic significance, SMEs continue to face persistent financing constraints, a situation further exacerbated by global trade tensions. Prolonged capital turnover cycles, rising operational costs, and heightened cash flow pressures have collectively increased SMEs' reliance on external financing. Particularly, a defining characteristic of SMEs in China is their high degree of social embeddedness [12, 23]. These relationships, embedded within dense personal networks such as family ties, friendships, and long-term business partnerships, serve as critical channels for resource acquisition and risk mitigation and repositories of soft information [2, 36, 46], including informal reputations and relational constraints such as reciprocity obligations and implicit expectations of trust. Supply chain finance (SCF) has emerged as a promising financial innovation that leverages the credit strength of core enterprises to integrate upstream and downstream partners, enabling SMEs to access financing or credit enhancement through structured supply chain relationships [7, 11, 25]. By aligning financial flows with supply chain dynamics, SCF aims to enhance operational efficiency and strengthen the resilience of the supply chain system. However, SMEs located on the periphery of the supply chain often remain underserved by such mechanisms. Moreover, credit risks originating from these SMEs may propagate across the supply chain network, thereby threatening the stability of core enterprises and disrupting operational continuity [27, 40]. Therefore, effectively identifying and managing credit risks in SCF requires a deep understanding of the social embeddedness of SMEs and soft information contained within these networks. Leveraging these insights is essential for ensuring the stability and sustainable development of supply chains.

In complex business environments, such as supply chain finance, enterprise credit risk prediction faces unprecedented challenges. The continuous expansion of transaction data volumes, coupled with increasingly intricate inter-enterprise relationship networks and historical behavioral patterns, renders traditional risk assessment models inadequate owing to their simplistic structures and limited feature representation capabilities [11]. A critical limitation of existing models is their neglect of relational embeddedness, which captures essential informal dynamics such as reputational signals and implicit obligations. Constructing interpretable and generalizable risk modeling frameworks requires the capacity to integrate heterogeneous data sources and the ability to uncover latent relational structures and their temporal evolution. This is particularly important for modeling informal relational dynamics that influence SMEs credit behavior and for improving the robustness and reliability of credit risk predictions. In complex supply chain networks, SMEs often operate in highly dynamic and uncertain environments where credit profiles exhibit pronounced temporal patterns and volatility [44, 62]. Specifically, traditional machine learning approaches fail to adequately model intricate relational dependencies and latent risk contagion pathways among enterprises. Moreover, these methods often overlook the rich contextual signals embedded in highly heterogeneous and uncertain data, resulting in inadequate performance in terms of prediction accuracy, robustness, and generalization capability [40]. These shortcomings highlight the urgent need for more advanced modeling frameworks that can effectively identify and manage credit risks in SCF, particularly those that improve the precision of risk assessments, enhance the robustness of trust mechanisms, and accurately capture temporal dynamics.

In summary, the core advantage of SCF lies in its capacity to enhance systemic resilience through networked interconnectedness, which enables efficient resource allocation and improves enterprise risk assessment. The social embeddedness of SMEs is central to this interconnected structure. This concept serves as a micro-foundational



a Traditional cluster approach

b Traditional GCN approach

Fig. 1. (a) This figure shows that clustering based solely on financial indicators fails to distinguish risk categories on the ECAD dataset, neglecting critical non-financial factors like legal disputes. (b) This figure shows that GCN-based representations fail to distinguish risk categories on the ECAD dataset, struggling to capture complex relational and risk-relevant features.

mechanism that shapes relational trust, facilitates soft information exchange, and imposes informal constraints on credit behavior. However, the prevailing credit risk evaluation paradigms exhibit fundamental misalignment. This misalignment stems from three key limitations: isolation (focusing on atomized enterprises), staticity (relying on historical snapshots), and neglecting the social embeddedness dimension. These characteristics undermine the applicability of current approaches in the inherently dynamic and networked context of the SCF. For instance, when evaluating an enterprise's creditworthiness, relying solely on conventional indicators, such as financial statements, while overlooking potential negative events involving upstream and downstream partners (e.g., legal disputes and default records), may lead to insufficient risk identification and compromise the effectiveness of risk mitigation strategies. Furthermore, within complex and evolving supply chain networks, an enterprise's credit status demonstrates pronounced temporal dynamics shaped by external environmental factors and interdependent behaviors across multiple stakeholders. This fundamental misalignment gives rise to a core challenge in supply chain finance credit risk management, manifested in three critical shortcomings: *semantic disentanglement deficiency in risk feature extraction*, *trust-aware modeling limitation in networked risk propagation*, and *spatio-temporal simulation constraint in dynamic risk contagion*. These limitations compromise the stability and practical applicability of existing credit risk prediction platforms, highlighting the need for more comprehensive and adaptive modeling approaches. A key strategy to address this paradigm misalignment is to integrate SMEs' social embeddedness as a core risk dimension, effectively leveraging the embedded soft information and behavioral norms. This integration enables a more holistic and dynamic assessment of supply chain risk, as elaborated below.

- ***CH1****: Semantic Disentanglement Deficiency in Risk Feature Extraction.* In SCF, an enterprise's credit risk arises not only from its internal operational conditions but also from its embeddedness within a complex supply chain network and dynamic temporal interactions over time [24, 31, 34]. However, conventional approaches relying on static financial indicators fail to capture clustering risk patterns, as demonstrated by both SMEsD and ECAD datasets: the supply chain networks exhibit strong community structures (SMEsD: $Q = 0.9635$, ECAD: $Q = 0.6191$) with bankruptcy risk clustering within communities (SMEsD: $H_{risk} = 0.4017$, ECAD: $H_{risk} = 0.9655$), yet traditional methods show near-zero correlation with these patterns (SMEsD: NMI=0.0004, ECAD: NMI=0.0029). Specifically, these methods frequently ignore critical non-financial indicators (e.g., legal disputes, fulfillment records, and partnership stability) and fail to capture the latent semantic structures and clustering patterns that underlie these features in terms of risk propagation mechanisms, as shown in Fig. 1(a). More importantly, such approaches typically model enterprises in isolation, disregarding their network positions, interaction dynamics, and systemic influence within the broader supply chain ecosystem [37]. Consequently, the key dimensions of interconnected risk remain undetected, undermining the accuracy and robustness of credit risk assessment systems.

- ***CH2***: *Trust-Aware Modeling Limitation in Networked Risk Propagation.* Trust underpins stable collaboration in SCF [32, 55]. A credit crisis at a core enterprise may rapidly propagate through upstream and downstream relationships, triggering a "chain reaction" that escalates into systemic risk [14, 32]. However, financial institutions lack the capability to perform graph-based granular trust assessments, limiting their ability to distinguish relationship quality. Traditional risk propagation models often fail to infer inter-enterprise trust levels from historical interactions and neglect to incorporate trust as a critical constraint or transmission weight in the modeling of risk diffusion [19]. This oversight leads to inaccurate simulations of risk spread across networks and diminishes the ability to distinguish between different risk categories, as illustrated in Fig 1(b). Without effective analysis of relational networks and transactional histories, establishing reliable trust evaluation frameworks remains challenging, hindering trust formation between capital providers and borrowers and increasing the likelihood of internal trust erosion, which undermines supply chain stability.
- ***CH3***: *Spatio-Temporal Disintegration in Dynamic Propagation Modeling*. The aforementioned limitations converge into a fundamental challenge: traditional models lack the capacity to capture the dynamic propagation of credit risk in supply chain networks. In complex systems such as SCF, enterprise credit risk is shaped not only by internal operational factors but also by inter-enterprise interactions that evolve over time and across network structures [6, 18, 23, 59]. Existing approaches struggle to jointly model the temporal state evolution (e.g., cyclical fluctuations) and spatial pathways of risk contagion. Temporal models, such as RNNs, effectively capture individual-level dynamics but neglect the topological dependencies between enterprises. Conversely, graph-based models, such as GNNs, encode structural relationships but fail to represent long-term, multi-scale temporal patterns. This spatiotemporal disintegration prevents accurate modeling of risk diffusion paths driven by both temporal shifts and network configurations, ultimately undermining the reliability and precision of enterprise credit risk assessments.

These limitations stem from an assessment paradigm that fails to account for the intrinsic interconnectedness of supply chain systems. This misalignment not only undermines the accuracy of risk predictions but may also result in systemic distortions in enterprise-level risk management strategies. In response to these challenges, we propose TrustVAE, a novel end-to-end generative framework that inherently integrates iterative clustering representation, a trust-aware transmission mechanism, and spatiotemporal modeling into a unified paradigm for enterprise risk assessment. Our research underscores that mitigating credit risks in SCF is not solely a technical endeavor. It requires a policy-informed and management-driven framework that aligns theoretical insights, practical implementation, and strategic coordination. Specifically, for **CH1**, we propose the Iterative Latent Risk Aggregation (ILRA) Module, which leverages deep nonlinear transformations and iterative clustering to unsupervised learn latent risk representations embedded with interpretable structured features from raw enterprise data. This module establishes a foundational and semantically enriched feature space for downstream modeling. In response to **CH2**, we propose the Bounded Graph Attention Network (BGAN) module, which introduces a trust-aware mechanism as bounded confidence on risk propagation. The BGAN dynamically quantifies the latent trust between enterprises using transaction histories and negative events, incorporating this trust intensity as a prior constraint in the graph attention computation. This study presents the first attempt to integrate trust into risk propagation modeling within GNNs, enabling explicit suppression of risk diffusion along low-trust edges, thereby improving both the semantic interpretability and structural robustness of inter-enterprise credit networks. For **CH3**, we further develop the Temporal Graph Autoregressive Flow (TGAF) module, which innovatively integrates temporal dependencies with autoregressive flow modeling to characterize credit risk evolution across multi-scale temporal horizons (e.g., daily, weekly, and monthly). This advancement substantially improved the model's ability to capture dynamic propagation effects and temporal evolutionary patterns. Extensive experiments demonstrated that TrustVAE consistently outperformed existing baseline methods in terms of prediction accuracy, robustness, and adaptability across varying data conditions.

In summary, our study systematically addresses the major challenges outlined above, marking a paradigm shift from isolated, static, and unidimensional enterprise credit risk prediction in supply chain finance to a dynamic, trust-aware modeling framework. The core contributions are as follows:

- We propose TrustVAE, a novel end-to-end generative paradigm that inherently integrates iterative clustering node aggregation, trust-aware edge propagation, and spatial-temporal graph autoregression into a unified architecture tailored for enterprise credit risk assessment in SCFs.
- An unsupervised ILRA module is proposed to learn intrinsically structured node aggregation through deep nonlinear transformations and iterative clustering. It can effectively capture latent risk patterns beyond static financial indicators and enhance structural expressiveness in graph-based risk modeling.
- A BGAN module is introduced to integrate implicit trust as bounded confidence into edge propagation, dynamically modulating risk diffusion by suppressing transmission across low-trust enterprise connections and thereby improving the fidelity and robustness of relational risk assessment in supply chain networks.
- A novel TGAF module is designed for coupling autoregressive flow modeling with temporal graph networks to jointly capture multi-scale temporal patterns and graph-structured dependencies, enabling a unified and expressive characterization of spatio-temporal risk evolution beyond the limitations of conventional RNNs or GNNs-based paradigm.

The remainder of this paper is organized as follows: Section 2 reviews the related literature. Section 3 presents the background and outlines the research questions. Section 4 describes the proposed method. Section 5 describes the experiments, analyzes the results, and discusses the findings. Section 6 concludes the paper and suggests future directions.

2 Related work

2.1 Traditional methods for financial risk prediction

Financial risk prediction is a critical component of credit portfolio management, decision-making, and financial system stability [28]. Traditional methodologies for credit risk assessment include classical credit scoring models, financial ratio analysis, and statistical inference techniques. These approaches vary in their assumptions and applicability but often exhibit limitations in terms of generalizability and adaptability to complex, real-world scenarios. Credit scoring models, which typically rely on historical credit data, demographic characteristics, and behavioral indicators, are widely used to assess creditworthiness and determine lending terms such as interest rates [35]. These models can be tailored to specific industries (e.g., mortgage finance) or customized according to lenders' internal risk preferences [1], although they generally lack flexibility to capture nonlinear patterns in borrower behavior. In recent years, machine learning techniques have gained increasing attention in credit risk modeling. Supervised learning algorithms such as Logistic Regression [22], Decision Trees [9], and ensemble algorithms like XGBoost and LightGBM have demonstrated improved predictive accuracy over classical models [47]. However, most of these models operate under the assumption of feature independence and are not inherently designed to model the relational or networked structures among entities. To address this limitation, some studies have explored the integration of graph-based features into credit risk modeling to capture inter-entity dependencies [13]. However, many of these methods rely on predefined graph structures that may fail to reflect the dynamic and evolving relationships in real-world financial systems. This challenge is particularly evident in SCF, where stakeholders such as suppliers, manufacturers, logistics providers, and financiers are connected through intricate transactional and operational linkages [50]. The complexity of such systems arises not only from the number of participating entities but also from nonlinear and time-varying interactions between them. Traditional machine learning models are generally inadequate in capturing these high-dimensional relational patterns, resulting in suboptimal performance in identifying latent credit risk signals, especially for SMEs, which are central to many supply chain ecosystems.

2.2 Graph-based methods for financial risk prediction

A major shortcoming of conventional machine learning lies in treating enterprises as isolated entities, ignoring the relational structure embedded within enterprise networks. This leads to performance degradation in scenarios involving sparse data or coordinated fraud [50]. Graph-based methods address this by explicitly modeling the interdependencies among financial actors. Graph Neural Networks (GNNs) enable joint learning over node attributes and relational structures through message passing across connected entities [49], introducing relational inductive biases aligned with transactional, ownership, or supply chain dynamics [3]. By aggregating and transforming neighbor information, GNNs capture latent interdependencies and risk propagation pathways that are challenging for traditional models to detect [10]. Recent studies have highlighted several effective graph-based architectures. Wang et al. developed a Relational Graph Attention Network (RGAT) to address SMEs data scarcity by capturing topological patterns [50]. Wei et al. combined Heterogeneous Hyper-Graph Neural Networks (HHGNN) with HGTN for improved bankruptcy prediction [53], while GDAN leverages dimension-level attention to enhance representation learning and has shown strong empirical performance [52]. Despite their advantages, most current approaches overlook the role of trust in shaping the relationship strength and information flow. Trust influences how risks propagate, how neighbors are weighted, and how network structures are interpreted [41]. To address this gap, we propose BGAN, a novel graph-based module with a trust-aware mechanism. Unlike standard GNNs that focus solely on structural connectivity, BGAN can adjust information aggregation based on the trust levels between nodes, distinguishing reliable from unreliable neighbors and improving the fidelity of risk propagation pathways. From an information systems perspective, the BGAN better captures the nuanced interactions among financial entities, enabling more accurate risk assessments in complex, interconnected environments.

2.3 VAE-based methods for financial risk prediction

VAEs have gained increasing attention as powerful generative modeling tools for financial risk prediction, particularly in settings involving sparse, high-dimensional, or incomplete data [4, 56]. By learning probabilistic latent representations, VAEs can uncover hidden risk factors and uncertainty distributions in enterprise data [16]. Several recent studies have demonstrated the effectiveness of VAEs in enterprise credit modeling. Xiao et al. proposed a VAE–DF, which integrates a VAE with a deep forest classifier to generate synthetic minority-class samples, improving the classification performance on imbalanced Internet finance datasets [54]. Huang et al. developed VDFDP, which uses view-specific encoders and attention-based fusion to learn unified latent representations from incomplete multi-view enterprise data, outperforming existing financial distress prediction methods [17]. Further advancing the field, Muthukumaran et al. proposed MVOFS-OVAE, a framework that combines multi-objective optimization with variational autoencoder-based classification to discover discriminative features and uncover latent risk patterns, thereby achieving state-of-the-art performance [33]. Tan et al. proposed DPTVAE, a conditional generative model for synthesizing realistic credit-bureau data using adaptive priors and hybrid decoding strategies, supporting both model training and privacy-preserving data sharing [45]. These studies highlight the distinct advantages of VAEs in uncovering meaningful latent structures from noisy or incomplete financial datasets. By modeling uncertainty and generating synthetic samples, VAEs enhance the generative capability and robustness of enterprise risk models. These strengths are critical in real-world applications, where data completeness and quality are often limited.

3 Preliminaries

3.1 Basic definitions

Definition 3.1 (Multi-relational Heterogeneous Enterprise Graph). In the context of SCF, a *multi-relational heterogeneous enterprise graph* is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where

- $\mathcal{V}$ denotes a set of nodes representing enterprises or associated entities.
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$ is a set of edges, each representing a specific relationship between two entities.
- $\mathcal{R}$ is a finite set of relationship types with $|\mathcal{R}| > 2$.
- A mapping function $\phi : \mathcal{E} \rightarrow \mathcal{R}$ assigns each edge to a specific relationship type.

This structure enables the representation of diverse and complex inter-enterprise relationships in financial networks.

Definition 3.2 (Enterprise Credit Risk Assessment). Given a set of multi-source heterogeneous risk data for enterprises, including the basic information set $\mathbf{H}$ and the multi-relational heterogeneous enterprise graph $\mathcal{G}$, the objective is to assess the credit risk level of enterprises by jointly modeling their intrinsic risks and contagion risks propagated through inter-enterprise networks. *Enterprise credit risk assessment* consists of the following steps:

- *Collection and Processing.* Enterprise-level intelligence is collected and preprocessed from multiple data sources into the feature set $\mathbf{H}$, including financial statements, market performance indicators, and legal dispute records.
- *Graph Construction.* Construct the multi-relational heterogeneous enterprise graph $\mathcal{G}$ based on the observed inter-enterprise relationships.
- *Risk Modeling.* Appropriate algorithms are applied to propagate information over $\mathcal{G}$, simulating the diffusion of risk across the network and generating a comprehensive credit risk score for each enterprise.

3.2 Problem formulation

The goal is to learn a predictive model f that maps the feature representation of an enterprise v to its credit risk score, denoted as $f : \mathcal{V} \rightarrow [0, 1]$. The model is trained on a dataset $\mathcal{D}$ that combines both the enterprise's basic information set $\mathbf{H}$ and its multi-relational heterogeneous graph structure $\mathcal{G}$, with the objective of minimizing the prediction error. Model performance is typically evaluated using metrics such as the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

Formally, given dataset $\mathcal{D} = \{(v_i, y_i)\}_{i=1}^{N}$, where $v_i \in \mathcal{V}$ represents the feature vector of the i-th enterprise, derived from both its basic information $\mathbf{H}$ and its position in the multi-relational heterogeneous graph $\mathcal{G}$. $y_i \in \mathcal{Y}$ denotes the ground truth credit risk label of the i-th enterprise (e.g., $y_i = 1$ for high-risk and $y_i = 0$ for low-risk). The goal is to find a function $f : \mathcal{V} \rightarrow [0, 1]$ such that $\hat{y}_i = f(x_i)$, where $\hat{y}_i$ is the predicted credit risk score for the i-th enterprise. The model f should be optimized to minimize the prediction error on $\mathcal{D}$. Specifically:

- *Nodes Information Aggregation.* Each enterprise's node vector v integrates both its internal features (captured in $\mathbf{H}$) and its relational context derived from the multi-relational heterogeneous graph $\mathcal{G}$.
- *Edges Information Propagation.* The model propagates risk signals across the edge network of $\mathcal{G}$, capturing both direct and indirect influences on the enterprise credit risk.
- *Prediction Objective.* Model f is optimized to minimize the prediction error between the estimated credit risk scores $\hat{y}_i$ and the true labels y_i across all enterprises in the dataset $\mathcal{D}$.

3.3 Variational Autoencoder

In the context of graph learning, we employed VAEs to encode input graph structures into a continuous latent space using an encoder and reconstruct them using a decoder. VAEs are unsupervised deep generative models composed of two core components: an encoder that maps observed data to a latent distribution and a decoder that reconstructs the original input from this latent representation.

Specifically, given a multi-relational heterogeneous enterprise graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, represented by node tensor $\mathbf{V}$, edge attribute tensor $\mathbf{E}$, and edge type tensor $\mathbf{R}$, our objective is to learn an encoder-decoder pair that maps the graph representation $\mathbf{G}$ to its latent embedding $\mathbf{z} \in \mathbb{R}^D$. Within the probabilistic framework of VAEs:

- The encoder models the variational posterior $q_\phi(\mathbf{z}|\mathbf{G})$,
- The decoder characterizes the generative distribution $p_\theta(\mathbf{G}|\mathbf{z})$,
- A prior $p(\mathbf{z})$ is imposed to regularize the latent representation.

Here, ϕ and θ denote the learnable parameters of the encoder and decoder. We adopt a standard isotropic Gaussian prior $p(\mathbf{z}) = \mathcal{N}(0, I)$ for the sake of simplicity and stability.

The VAE objective function balances two components:

$$\mathcal{L}_{\text{VAE}}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{G})}[-\log p_\theta(\mathbf{G}|\mathbf{z})] + \mathbb{D}_{\text{KL}}[q_\phi(\mathbf{z}|\mathbf{G}) \| p(\mathbf{z})], \tag{1}$$

where the first term encourages the accurate reconstruction of the input graph and the second term ensures that the latent distribution remains close to the prior. This formulation enables us to effectively capture complex structural patterns in enterprise networks while modeling risk propagation across the supply chain finance system.

4 The TrustVAE Approach

4.1 Overview

As illustrated in Fig. 2, we propose an end-to-end generative paradigm to enhance enterprise credit risk prediction. The framework comprises three core components: **(i)** *Iterative Latent Risk Aggregation*, designed to address **CH1** through deep nonlinear transformations and iterative clustering, discovering latent structural patterns to support effective node information aggregation. **(ii)** *Bounded Graph Attention Network* alleviates the **CH2**, which incorporates bounded confidence constraints into graph-based risk propagation to better capture realistic patterns among enterprises. **(iii)** *Temporal Graph Autoregressive Flow* addresses the issue of **CH3**, which models dynamic evolution of credit risk across both temporal and topological dimensions in supply chain networks by integrating multi-scale temporal dynamics ranging from daily to monthly horizons. These components are jointly optimized within a unified architecture, enabling the method to capture complex risk patterns while improving its predictive accuracy and robustness.

4.2 Enterprise information projection

This section introduces *enterprise information projection*, which maps raw enterprise node attributes, including basic enterprise characteristics and litigation information [53], into a latent vector space and extracts risk-relevant features. The resulting representation is denoted as $\mathbf{H} \in \mathbb{R}^{N \times d}$, where N is the number of enterprises, and d is the dimensionality of the latent space. The input features were initially linearly projected into a shared latent space. Then, a normalization operation is applied to stabilize the node representation during training. Finally, an activation function is used to introduce nonlinearity into the learned representations. The specific implementation is:

$$\mathbf{H'} = \sigma(\text{Norm}(\mathbf{HW} + \mathbf{b}^\top)), \tag{2}$$

where $\mathbf{W}$ and $\mathbf{b}$ are learnable parameter matrices, σ is the activation function, and $\mathbf{H'}$ represents the new node representation after mapping.

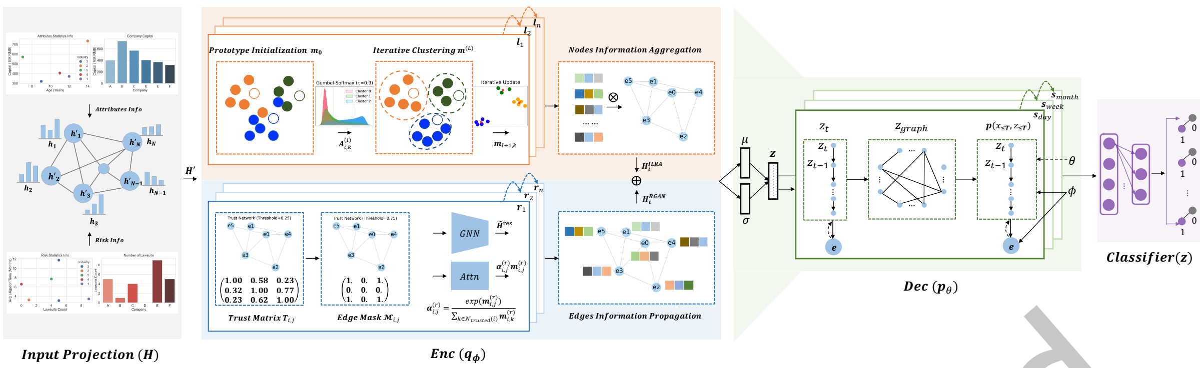


Fig. 2. Overview of TrustVAE. In the diagram, the orange section represents the ILRA module, the blue section denotes the BGAN, the green section corresponds to the TGAF module, and the purple section indicates the Integration and Joint Prediction Module.

4.3 Iterative latent risk aggregation module

Extracting informative risk representations from heterogeneous enterprise data remains a core challenge in credit risk prediction, particularly in SCF. Existing models, such as MacridVAE [29], ComiRec-SA [5], VAEneu [21], and MulVAEK [24] have demonstrated effectiveness in handling static or sequential data. However, these methods often rely on simplified assumptions and lack the capacity to model evolving risk patterns embedded in relational networks. To address these limitations, we propose the ILRA module, which employs an iterative clustering mechanism to refine node representations through risk information aggregation within enterprise networks. By jointly optimizing cluster assignments and node embeddings across iterations, ILRA captures the higher-order structural dependencies that characterize complex risk propagation dynamics. As elaborated in Section 5.4.2, this formulation enhances the representational capacity of credit risk modeling in supply chain ecosystems. The learned representations seamlessly integrate with downstream temporal and topological modeling components to facilitate coherent and context-aware risk inference.

4.3.1 Cluster prototype initialization. As depicted in the orange section of Fig. 2, the ILRA module integrates three core components: *Cluster Prototype Initialization*, *Iterative Clustering Process*, and *Nodes Information Aggregation*. The overall framework is implemented as a neural network module with configurable hyperparameters, including the latent dimension d, number of clusters K, and number of iterations L.

The process begins with the initialization of the cluster prototypes in the latent space. Specifically, K initial prototypes $\mathbf{m}_0 \in \mathbb{R}^{K \times d}$ are sampled from a Gaussian distribution as follows :

$$\mathbf{m}_0 \sim \mathcal{N}(0, I_d), \tag{3}$$

where I_d denotes the $d \times d$ identity matrix. To facilitate clustering, the input node representations $\mathbf{H'}$ are first projected into the shared latent space through a linear transformation as follows:

$$\tilde{\mathbf{H}} = \text{Linear}(\mathbf{H'}) = \mathbf{H'W} + \mathbf{b}, \tag{4}$$

where $\mathbf{W} \in \mathbb{R}^{D \times d}$ and $\mathbf{b} \in \mathbb{R}^d$ denote the learnable weight matrix and bias vector, respectively.

4.3.2 Iterative clustering process. Following initialization, the ILRA performs iterative refinement of cluster assignments and prototypes over L steps. This process consists of three sequential stages: similarity-based assignment, prototype update, and iteration termination. At each iteration l, the similarity between each sample $\tilde{\mathbf{H}}_i$ and prototype $\mathbf{m}_{i,k}$ is computed using the cosine distance, normalized by a temperature parameter τ to control the sharpness of the resulting distribution:

$$\text{sim}_{i,k}^{(l)} = \frac{\tilde{\mathbf{H}}_i^\top \mathbf{m}_{l,k}}{\tau \cdot \|\tilde{\mathbf{H}}_i\|_2 \cdot \|\mathbf{m}_{l,k}\|_2}. \tag{5}$$

A Gumbel-Softmax mechanism is then applied to introduce stochasticity while enabling end-to-end differentiability as follows:

$$\mathbf{A}_{i,k}^{(l)} = \frac{\exp\left(\frac{\text{sim}_{i,k}^{(l)} + g_i^{(l)}}{\tau}\right)}{\sum_{k'=1}^{K} \exp\left(\frac{\text{sim}_{i,k'}^{(l)} + g_i^{(l)}}{\tau}\right)}, \tag{6}$$

where $g_i^{(l)} \sim \text{Gumbel}(0, 1)$. The resulting soft assignment matrix $\mathbf{A}^{(l)} \in \mathbb{R}^{N \times K}$ captures the probabilistic membership of each sample across all clusters. Using the current assignment matrix $\mathbf{A}^{(l)}$, prototypes are updated to better reflect the underlying data structure:

$$\mathbf{m}_{l+1,k} = \frac{\sum_{i=1}^{N} \mathbf{A}_{i,k}^{(l)} \cdot \tilde{\mathbf{H}}_i}{\sum_{i=1}^{N} \mathbf{A}_{i,k}^{(l)} + \epsilon}, \tag{7}$$

where ϵ is a small constant (e.g., 10^{-8}) to prevent division by zero. This iterative update ensures that prototypes gradually converge to stable representations of latent risk clusters.

The clustering process is repeated for L iterations or until convergence. The final output includes the refined prototypes $\mathbf{m}^{(L)}$ and the corresponding assignment matrix $\mathbf{A}^{(L)}$, which serve as the basis for downstream feature aggregation.

4.3.3 Nodes information aggregation. With the final cluster assignments $\mathbf{A}^{(L)}$, ILRA aggregates the input node representations $\mathbf{H'}$ across samples to form cluster-level representations. For each cluster k, the aggregated feature vector is computed as:

$$\mathbf{H'}_k = \sum_{i=1}^{N} \mathbf{A}_{i,k}^{(L)} \cdot \mathbf{H'}_i, \tag{8}$$

where $\mathbf{H'}_k \in \mathbb{R}^D$ denotes the aggregated feature vector for cluster k. This operation effectively transforms the node-level features into cluster-level representations by leveraging the probabilistic cluster assignments. To ensure dimensional compatibility with BGAN's output, we apply a linear projection layer to align the dimensions:

$$\mathbf{H}_i^{\text{ILRA}} = \mathbf{W}_{\text{proj}} \cdot \mathbf{H'}_k, \tag{9}$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{D_{\text{BGAN}} \times D_{\text{ILRA}}}$ is a learnable weight matrix that projects $\mathbf{H}_i^{\text{ILRA}}$ to the same dimension D_{BGAN} as BGAN's output.

4.4 Bounded graph attention network module

In SCF, enterprise credit risk prediction faces increasing complexity owing to the dynamic inter-enterprise relationships and interactions. Traditional models struggle to capture fine-grained risk patterns owing to their limited representational capacity and oversimplified assumptions. While GNNs effectively model relational dependencies, they often overlook trust-based constraints on information propagation [15, 52, 57]. In contrast, VAE-based methods emphasize latent representation learning but lack mechanisms for capturing graph-level interactions [13, 21, 24, 58]. To address these limitations, we propose BGAN, a novel module that integrates graph message passing with bounded confidence to enable trust-aware risk propagation. Unlike conventional GNNs [8, 61, 63] that use cosine similarity for edge weights but ignore risk confidence bounds, BGAN introduces a similarity threshold to restrict message propagation to nodes whose risk profiles lie within a defined confidence bound. By embedding the bounded confidence mechanism within an attention framework, the BGAN selectively modulates message passing according to relational trust, thereby aligning risk propagation with empirically observed behavioral norms in supply chain ecosystems. This design enhances both the model interpretability and predictive accuracy, yielding a more robust alternative to existing GNN- and VAE-based approaches, as detailed in Section 5.4.3.

4.4.1 Trust matrix initialization. As illustrated in the blue component of Fig. 2, the BGAN module integrates three core components: *Trust Matrix Initialization*, *Trust-aware Edge Mask*, and *Edges Information Propagation*. The BGAN begins by initializing a similarity-based trust matrix $\mathbf{T} \in (0,1]^{N\times N}$ to quantify pairwise trust between nodes (e.g., enterprises). Given node embeddings $\mathbf{H'} \in \mathbb{R}^{N\times d}$, the Euclidean distance between nodes i and j is computed as:

$$D(i,j) = \|\mathbf{h'}_i - \mathbf{h'}_j\|_2. \tag{10}$$

Rather than applying a hard thresholding mechanism, the trust score between nodes is modeled as a smooth inverse function of distance:

$$T_{i,j} = \frac{1}{1 + D(i,j)}. \tag{11}$$

This formulation ensures that closer nodes in the embedding space receive higher trust scores, with values naturally bounded in $(0,1]$. Self-confidence is enforced by setting the diagonal entries to 1:

$$T_{i,i} = 1, \quad \forall i \in \{1,\ldots,N\}. \tag{12}$$

This trust matrix serves as a continuous, semantically interpretable measure of node relationships, forming the basis for subsequent edge filtering and message passing.

4.4.2 Trust-aware edge mask. The trust-aware edge mask constitutes a fundamental component of the BGAN module, designed to capture the nuanced relationship quality between enterpirses in the supply chain, addressing a critical gap in existing graph-based risk propagation models. As established in Section 4.4.1, we derive trust scores from node embeddings using Eq. 11, where $\mathbf{h'}_i$ and $\mathbf{h'}_j$ represent the embedding vectors of enterpirses i and j. This formulation ensures trust scores are bounded within $(0,1]$, with higher values indicating stronger trust relationships. To filter low-trust relationships while preserving meaningful connections, we introduce a trust-aware edge mask with a threshold $\gamma \in (0,1]$. The binary mask is defined as:

$$\mathcal{M}_{i,j} = \begin{cases} 1, & \text{if } T_{i,j} > \gamma \\ 0, & \text{otherwise} \end{cases}. \tag{13}$$

The filtered edge set $\mathcal{E}_{\text{trusted}}$ is then given by:

$$\mathcal{E}_{\text{trusted}} = \{(i,j) \in \mathcal{E} \mid \mathcal{M}_{i,j} = 1\}. \tag{14}$$

Theoretical Justification: This mechanism is grounded in the theoretical framework of social embeddedness in supply chain finance, where trust serves as a core determinant of financial stability. High-trust relationships (e.g., long-term partners with consistent transaction records) exhibit strong risk propagation, as financial distress in one partner is likely to impact trusted counterparts due to deep integration and mutual dependency. Conversely, low-trust relationships (e.g., one-time transactions or enterpirses with legal disputes) demonstrate weak risk propagation, as these connections lack the stability and transparency for risk to spread effectively. The trust-aware edge mask explicitly models this distinction, which is absent in traditional graph-based risk propagation approaches that treat all edges equally.

The threshold γ is determined through empirical analysis on the validation set, balancing the trade-off between preserving meaningful relationships and filtering out noise. We conducted extensive experiments across a range of γ values (from 0.1 to 0.9 in 0.1 increments) and selected $\gamma = 0.5$ as the optimal value, which maximized the AR metric on our validation set. This value was consistent across both datasets, indicating its robustness across different supply chain structures. This trust-aware mechanism aligns with the theoretical framework of social embeddedness in supply chain finance, where trust is a core determinant of financial stability. By dynamically computing trust scores from node embeddings, our approach captures the evolving nature of enterpirse relationships, unlike static trust metrics used in prior work. This dynamic modeling is particularly crucial in complex, evolving supply chains where trust relationships change over time.

4.4.3 Edges information propagation. The information aggregation process combines GNN-based message passing with relation-aware attention mechanisms to model heterogeneous risk propagation with the explicit integration of a trust-aware edge mask. Given the filtered edge set $\mathcal{E}_{\text{trusted}}$, the network performs the following steps.

1) Trust-aware edge filtering. The trust mask $\mathcal{M}$ is applied to the original edge index and edge type tensors as follows:

$$\mathcal{E}_{\text{trusted}} = \{(i,j) \in \mathcal{E} \mid \mathcal{M}_{i,j} = 1\}, \quad \mathcal{T}_{\text{trusted}} = \{r \in \mathcal{T} \mid (i,j) \in \mathcal{E}_{\text{trusted}}\}. \tag{15}$$

This ensures that subsequent operations (e.g., normalization and message passing) are computed only on the trusted subgraph.

2) Residual connection with masked aggregation. For the filtered edge set $\mathcal{E}_{\text{trusted}}$, symmetric GNN normalization computes the edge weights as

$$\tilde{W}_{i,j} = \frac{1}{\sqrt{|\mathcal{N}_{\text{trusted}}(i)| \cdot |\mathcal{N}_{\text{trusted}}(j)|}}, \tag{16}$$

where $|\mathcal{N}_{\text{trusted}}(k)|$ denotes the degree of node k in trusted subgraph. This normalization scales neighbor contributions based on local connectivity patterns. A residual connection combines input projections with normalized aggregation over the trusted subgraph as follows:

$$\tilde{\mathbf{h}_i^{\text{res}}} = \mathbf{W}_{\text{input}}(\mathbf{h'}_i) + \sum_{j \in \mathcal{N}_{\text{trusted}}(i)} \tilde{W}_{i,j} \cdot \mathbf{h'}_j, \tag{17}$$

where $\mathbf{W}_{\text{input}}$ is a learnable linear projection matrix, $\mathcal{N}_{\text{trusted}}(i)$ denotes the neighbors of node i in the trusted subgraph, and the aggregation uses filtered edge weights $\tilde{W}_{i,j}$ to ensure robustness against spurious connections.

3) Relation-specific attention mechanism. For each edge type $r \in \{1, \dots, R\}$ in $\mathcal{E}_{\text{trusted}}$, distinct linear transformations $\mathbf{W}_{\text{rel}}^{(r)} \in \mathbb{R}^{d \times d}$ are applied to compute messages:

$$\mathbf{m}_{i,j}^{(r)} = \text{ReLU}(\mathbf{W}_{\text{rel}}^{(r)} \mathbf{h'}_j). \tag{18}$$

Relation-specific attention coefficients $\alpha_{i,j}^{(r)}$ are computed via softmax normalization over neighbors:

$$\alpha_{i,j}^{(r)} = \frac{\exp(\mathbf{m}_{i,j}^{(r)})}{\sum_{k \in \mathcal{N}_{\text{trusted}}(i)} \exp(\mathbf{m}_{i,k}^{(r)})}. \tag{19}$$

4) Final risk representation. The final risk representation aggregates both residual and attention-based messages.

$$\mathbf{H}_i^{\text{BGAN}} = \text{Norm}\left(\tilde{\mathbf{H}}^{\text{res}} + \sum_{r=1}^{R} \sum_{j \in \mathcal{N}_r^{\text{trusted}}(i)} \alpha_{i,j}^{(r)} \mathbf{m}_{i,j}^{(r)} \right), \tag{20}$$

where $\text{Norm}(\cdot)$ denotes layer normalization. This dual-path aggregation preserves both structural patterns and relation-specific dependency structures, enabling fine-grained risk propagation modeling in heterogeneous financial networks.

The final node representation is obtained by element-wise addition of ILRA and BGAN outputs:

$$\mathbf{H}_i^{\text{ALL}} = \mathbf{H}_i^{\text{ILRA}} + \mathbf{H}_i^{\text{BGAN}} \quad \forall i \in \{1, \dots, N\}. \tag{21}$$

This fusion strategy effectively combines cluster-level risk patterns with trust-aware edge propagation patterns, as shown in Fig. 2. The alignment step ensures compatibility between the two modules, while the element-wise addition preserves additive risk contributions from both perspectives.

4.5 Temporal graph autoregressive flow module

Enterprise credit risk in SCF is shaped by both temporal dynamics and evolving network interactions. However, existing methods often treat these aspects separately: recurrent models (e.g., RNNs) capture temporal dependencies but ignore topological structures, while GNNs encode relational patterns but lack the capacity to model long-term temporal evolution. This limitation hinders accurate modeling of risk contagion paths that depend on both time-varying behaviors and network interactions, particularly critical in SCF. To address this gap, we present TGAF, a principled module that jointly models multi-scale temporal dependencies and spatial dynamics within evolving enterprise networks. The TGAF constructs temporal graphs to represent inter-enterprise interactions over time and employs Gated Recurrent Units (GRUs) to capture both short-term fluctuations and long-term trends within this structure. By integrating temporal and topological information, the TGAF enables more precise and interpretable modeling of dynamic credit risk propagation in complex supply chain systems, as discussed in Section 5.4.4.

4.5.1 Latent variable sampling. In traditional VAE, the sampling process of the latent variables is achieved through a reparameterization trick. Specifically, given the mean μ and variance σ^2 from the encoder, the latent variable $\mathbf{z}$ is sampled using the reparameterization trick:

$$\mathbf{z} = \mu + \sigma \cdot \epsilon, \tag{22}$$

where $\epsilon \sim \mathcal{N}(0, 1)$. This ensures that $\mathbf{z}$ is distributed as $\mathcal{N}(\mu, \sigma^2)$. For latent variables, we can extend this to a sequence of latent variables $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_T$:

$$\mathbf{z}_t = \mu_t + \sigma_t \cdot \epsilon_t, \tag{23}$$

where μ_t and σ_t are functions of the previous latent variables $\mathbf{z}_{<t}$. This formulation ensures that $\mathbf{z}_t$ captures the temporal dependencies from previous time steps.

4.5.2 Latent temporal dependencies. As shown in the green component of Fig. 2, the TGAF module integrates three core components: *Latent Temporal Dependencies*, *Latent Temporal Graph Construction*, *Encoding of Entire Latent Temporal Dependencies*, and *Multi-Scale Temporal Aggregation*. Latent temporal dependencies refer to the interdependencies among latent variables across different time steps in time series data. In our SCF context, these dependencies capture how credit risk evolves over time within the supply chain network. To capture these dependencies, we introduced a temporal prior distribution that explicitly models the autoregressive nature of credit risk dynamics. Suppose the latent variable $\mathbf{z}_t$ at time t depends on the state of the latent variable at the previous time step $t-1$, following an autoregressive process:

$$\mathbf{z}_t = f(\mathbf{z}_{t-1}) + \epsilon_t, \tag{24}$$

where $f(\mathbf{z}_{t-1})$ is a function representing the latent state at the current time step based on the previous state, and $\epsilon_t \sim \mathcal{N}(0, \sigma^2 I)$ is a Gaussian noise term. We model $f(\mathbf{z}_{t-1})$ as a linear transformation:

$$f(\mathbf{z}_{t-1}) = \mathbf{T}_s \mathbf{z}_{t-1} + \mathbf{b}, \tag{25}$$

where $\mathbf{T}_s$ is the state transition matrix and $\mathbf{b}$ is the bias vector.

Specifically, our temporal partitioning strategy employed in our dataset construction is critical for the effective operation of the TGAF module, as it ensures that the model learns time-dependent risk propagation patterns exclusively from historical data. Unlike the Ladder VAE [38, 43], which introduces a hierarchical structure for latent variables but does not model temporal dependencies, our approach explicitly conditions $\mathbf{z}_t$ on $\mathbf{z}_{t-1}$, capturing the dynamic nature of credit risk evolution. In the SCF context, this means that TGAF can effectively model how a company's credit risk at time t is influenced by its risk state at time $t-1$, as well as the risk propagation patterns within the supply chain network. This is crucial for accurately predicting credit risk in SCF, where risk can evolve and propagate through the network over time.

4.5.3 Latent temporal graph construction. To capture the evolving interdependencies among enterprises in SCF, we constructed a latent temporal graph that explicitly models how relational patterns change over time. Given a sequence of latent states $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_{T_s}$, we model their interdependencies via a learnable adjacency matrix $\mathbf{A}_t \in \mathbb{R}^{d_z \times d_z}$ for each time step t, where each element $A_{ij}^{(t)}$ represents the strength of influence from latent variable $\mathbf{z}_j$ to $\mathbf{z}_i$ at time t. Rather than predefined or binary connections, $\mathbf{A}_t$ is initialized as a parameter matrix and optimized during training to capture latent relational patterns dynamically based on temporal context:

$$\mathbf{A}_t = \sigma(\mathbf{W}_a \cdot \mathrm{MLP}(t, \mathbf{Z}_{\leq t})), \tag{26}$$

where $\mathbf{W}_a$ is a learnable weight matrix, t is the time index, and $\mathbf{Z}_{\leq t} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t]$ represents historical latent states. This formulation ensures that the relational structure adapts to both the current time step and the historical evolution of risk patterns.

The graph-enhanced representation $\mathbf{z}_t^{\mathrm{graph}}$ is then computed as:

$$\mathbf{z}_t^{\mathrm{graph}} = \mathbf{z}_t \mathbf{A}_t. \tag{27}$$

Unlike static adjacency matrices that assume fixed relationships, our formulation aligns with SCF reality where enterprise relationships continuously evolve. The dynamic nature of $\mathbf{A}_t$ ensures that the model captures temporal risk propagation patterns rather than fixed relational structures, making it particularly suitable for credit risk assessment in dynamic supply chain environments.

4.5.4 Multi-scale temporal aggregation. Given the sequence of graph-transformed latent features $\mathcal{Z}^{\mathrm{graph}} = \{\mathbf{z}_1^{\mathrm{graph}}, \ldots, \mathbf{z}_T^{\mathrm{graph}}\}$, we next model their temporal evolution using a GRU. Formally, the hidden state at time t is updated as follows:

$$\begin{aligned}
\mathbf{r}_t &= \sigma(\mathbf{W}_{rz}\mathbf{z}_t^{\mathrm{graph}} + \mathbf{b}_r), \\
\mathbf{u}_t &= \sigma(\mathbf{W}_{uz}\mathbf{z}_t^{\mathrm{graph}} + \mathbf{b}_u), \\
\tilde{\mathbf{h}}_t &= \tanh(\mathbf{W}_{hz}\mathbf{z}_t^{\mathrm{graph}} + \mathbf{W}_{hr}(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h), \\
\mathbf{h}_t &= (1 - \mathbf{u}_t) \odot \mathbf{h}_{t-1} + \mathbf{u}_t \odot \tilde{\mathbf{h}}_t,
\end{aligned} \tag{28}$$

where σ denotes the sigmoid function, and $\odot$ represents element-wise multiplication.

The temporal evolution of enterprise credit risk manifests across multiple granularities, including daily, weekly, and monthly dynamics. To effectively model such multi-scale temporal patterns, we adopted a multi-scale GRU architecture with parallel units operating at distinct temporal resolutions:

$$\mathbf{h}_t^{(s)} = \mathrm{GRU}^{(s)}(\mathbf{h}_t), \quad s \in \{\mathrm{day}, \mathrm{week}, \mathrm{month}\}. \tag{29}$$

Each GRU independently models the temporal dependencies at its respective scale. The outputs were concatenated and fused via a fully connected layer:

$$\mathbf{h}_t^{\text{fusion}} = \text{ReLU}\left(\mathbf{W}_{\text{fuse}} \begin{bmatrix} \mathbf{h}_t^{(\text{day})} \\ \mathbf{h}_t^{(\text{week})} \\ \mathbf{h}_t^{(\text{month})} \end{bmatrix} + \mathbf{b}_{\text{fuse}}\right), \tag{30}$$

yielding a unified representation that integrates information across different temporal resolutions. Finally, the refined latent representation $\mathbf{z}'_t$ is computed as:

$$\mathbf{z'}_t = \text{LayerNorm}(\mathbf{h}_t^{\text{fusion}}). \tag{31}$$

This formulation ensures that the updated latent codes capture both topological and temporal enterprise dynamics.

4.6 Integration and joint prediction module

To enable end-to-end credit risk prediction, we designed a joint learning framework that integrates disentangled risk representations with a downstream classification module, as shown in the purple component of Fig. 2. The classifier is implemented as a multi-layer perceptron (MLP) or any suitable neural architecture, denoted by $f(\cdot)$. This model maps the disentangled latent representation $\mathbf{z}$ to a predicted credit risk score $\hat{\mathbf{y}} \in [0,1]^N$. Formally, the prediction process is defined as:

$$\hat{\mathbf{y}} = f(\mathbf{z}), \tag{32}$$

where $\mathbf{z}$ is obtained from the ILRA and BGAN modules, respectively.

The overall training objective combined three complementary loss components: reconstruction loss, distribution regularization, and classification loss. The total loss function is defined as:

$$\mathcal{L}(\theta, \phi, \psi) = \mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{KLD}} + \mathcal{L}_{\text{BCE}}, \tag{33}$$

where:

- $\mathcal{L}_{\text{MSE}} = \sum_{i=1}^{N}(x_i - \hat{x}_i)^2$ ensures faithful reconstruction of input features.
- $\mathcal{L}_{\text{KLD}} = -0.5\sum_{i=1}^{N}\left(1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2\right)$ regularizes the latent space by aligning it with a prior Gaussian distribution.
- $\mathcal{L}_{\text{BCE}} = -\sum_{i=1}^{N}(y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i))$ optimizes the classification performance for binary credit risk prediction.

By jointly minimizing $\mathcal{L}$, the model learns both expressive latent representations and accurate risk prediction capabilities in a unified framework.

5 Experiments

In this section, we evaluate TrustVAE on two public benchmark datasets with distinct characteristics. We assessed its effectiveness, analyzed the contribution of each module, and addressed the following research questions (RQs):

- **RQ1**: How does TrustVAE perform in comparison to SOTA methods in credit risk prediction?
- **RQ2**: What is the individual contribution of each module in TrustVAE to overall performance?
- **RQ3**: How effectively does TrustVAE address the challenges outlined in Section 1 across different datasets?
- **RQ4**: How sensitive is TrustVAE's performance to variations in hyperparameter settings?

5.1 Experimental settings

5.1.1 Datasets. We evaluated the proposed model using two publicly available benchmark datasets, **SMEsD**[1] and **ECAD**[2], which are specifically designed for enterprise credit risk in SCF, as summarized in Table 1. These datasets provide rich, real-world representations of enterprise relationships and financial dynamics, directly reflecting the complexities of SCF where credit risk propagation through trust-based relationships is a critical concern.

- The **SMEsD** dataset comprises 3,976 Chinese small and medium enterprises (SMEs) operating in Zhejiang Province, one of China's most economically vibrant regions, covering the period from 2014 to 2021. This dataset was constructed to address the practical challenges of SCF risk assessment by capturing both structural and behavioral characteristics of enterprises within supply chain networks. Specifically, we identified 889 bankrupt SMEs as seed entities through official bankruptcy records, then systematically collected their business relationships (including shareholder, executive, and investment connections) from public sources. The dataset includes two node types (enterprises and individuals) and five relationship types (executive, shareholder, other stakeholders, inter-enterprise shareholder, investment-holding, and branch relationships), with investment-holding relationships being weighted by equity proportion. For feature construction, we extracted 32 meaningful features from enterprise operational and legal histories: (1) basic enterprise information (registered capital, paid capital, establishment date); (2) litigation records (32 features including plaintiff/defendant roles, case causes, court levels, verdict outcomes, and timestamps); and (3) relationship-based features (e.g., number of connected enterprises, relationship strength metrics). Each enterprise was labeled based on bankruptcy status (Bankrupt/Survive) as of its last observation date, with the dataset temporally partitioned into training, validation, and test sets using strict chronological cutoffs to preserve temporal integrity. Specifically, we employed a temporal split where the earliest 60% of the time period formed the training set, the subsequent 20% constituted the validation set, and the latest 20% comprised the test set.
- The **ECAD** dataset represents a large-scale enterprise network with 48,758 companies and over one million interconnections, covering the period from 2017 to 2022. This dataset was specifically designed to model enterprise credit risk propagation in complex supply chains, with features directly aligned with SCF risk management practices. The construction process followed rigorous methodology: we selected 512 enterprises with valid credit ratings as seed entities, then identified their one-hop neighbors across three relationship types (investor-holders, branches, shareholders), resulting in a comprehensive network reflecting typical SCF relationships. Feature engineering was carefully designed for SCF context: (1) basic enterprise features (registered capital, paid capital, establishment date, administrative awards, penalties, licenses); (2) litigation features (12 cause types, 2 role types, and 5 temporal categories); and (3) relationship features (e.g., number of connections, relationship duration, trust scores). All features were standardized and converted to numerical representations for model training. The dataset was partitioned into training, validation, and test sets based on credit rating observation dates using strict chronological cutoffs, with the earliest 60% of the time period forming the training set, the subsequent 20% constituting the validation set, and the latest 20% comprising the test set. Each enterprise was labeled according to its credit rating status (Good: AAA, Normal: AA, Bad: A, BBB, BB, etc.), with this feature design directly supporting SCF credit risk assessment by capturing both financial health indicators and relationship-based risk factors critical for supply chain credit management.

[1]https://github.com/shaopengw/ComRisk
[2]https://github.com/shaopengw/GDAN

Table 1. Statistics of dataset after preprocessing

Datasets			Training	Validation	Testing
SMEsD	#Node		3976		
	#Relation	#Manager	3944		
		#Shareholder	5722		
		#Stakeholder	6402		
		#Branch	768		
		#Investor holder	9408		
	#Label	#Bankrupt	1621	354	318
		#Survive	1195	367	173
ECAD	#Node		48758		
	#Relation	#Investor holder	77572		
		#Branch	12764		
		#Shareholder	1302415		
	#Label	#AAA	327	122	117
		#AA	378	108	91
		#Others	74	28	29

5.1.2 Evaluation protocol and metrics. For a rigorous evaluation of our proposed model, we adopted widely recognized performance metrics from prior studies [30, 60], including the Area Under the Receiver Operating Characteristic Curve (ROC-AUC), Accuracy Ratio (AR), Accuracy, and F1 score. The ROC-AUC was selected as the primary metric because of its effectiveness in assessing ranking capability, with the other metrics providing supplementary insights. Credit scoring models must prioritize high-risk clients based on probability values, which serve as a natural basis for ranking [30, 42]. Meanwhile, AR reflects the model's discriminatory ability, with default probabilities providing actionable insights for banks to develop targeted strategies. Furthermore, to assess the classification performance under class imbalance, we adopted the Accuracy and the F1 score. Accuracy reflects the overall proportion of correct predictions, while the F1 score, as the harmonic mean of precision and recall, provides a balanced evaluation by accounting for both false positives and false negatives. By utilizing these metrics, we ensured a thorough evaluation of the proposed solution's effectiveness and reliability in practical applications.

5.1.3 Comparison baselines. To evaluate the effectiveness of TrustVAE, we compared it with three categories of classification methods.

- *Traditional Approaches.* These include widely used machine learning algorithms such as Logistic Regression (**LogR**), Support Vector Machine (**SVM**), Decision Tree (**DT**), and Gradient Boosting Decision Tree (**GBDT**). These models are well established and serve as strong baselines for performance comparisons.
- *VAE-based Approaches.* This category focuses on variational autoencoder models applied to generative learning tasks. We consider **Vanilla VAE** [43], **Ladder VAE** [43], **Graph VAE** [13], **GPGVAE** [58], **VAEneu** [21], and **MulVAEK** [24].
- *Graph-Based Approaches.* We consider both homogeneous and heterogeneous graph neural network models. Homogeneous graph models include **GCN** [20] and **GAT** [48]. For heterogeneous graphs, we evaluate **RGCN** [39], **HAN** [51], **HGT** [15], **SeHGNN** [57], and **GDAN** [52]. These methods are particularly effective for modeling complex relational structures and latent variable interactions.

5.1.4 Hyperparameter settings. Within TrustVAE, multi-layer perceptrons (MLPs) are employed for both the encoder and decoder networks. A preliminary study was conducted to assess the impact of the MLP depth. The results indicated that a two-hidden-layer architecture with configurations {(64), (128, 64)} for the encoder yielded

Table 2. The performance comparison with baselines using the metrics of ROC-AUC (%) and AR (%), Accuracy (%), and F1 (%). The bold and underlined values denote the best and runner-up results, respectively.

Model	SMEsD				ECAD			
	ROC-AUC	AR	Accuracy	F1	ROC-AUC	AR	Accuracy	F1
LogR	67.26	34.52	76.99	62.95	67.67	35.34	60.34	42.63
SVM	52.21	4.42	65.38	10.53	65.65	31.30	60.76	42.17
DT	70.90	41.80	76.17	61.13	62.28	24.56	56.12	48.56
GBDT	83.00	66.00	74.13	47.74	77.84	55.68	61.60	54.98
Vanilla VAE	77.79	55.58	77.19	74.66	76.37	52.74	61.63	51.13
Ladder VAE	76.53	53.06	75.15	72.71	76.14	52.28	60.08	53.24
GraphVAE	80.69	61.38	76.58	73.72	75.80	51.60	63.18	52.88
GPGVAE	79.69	59.38	75.76	73.48	77.54	55.08	63.95	52.52
VAEneu	79.07	58.14	76.99	74.17	78.25	56.50	60.47	50.42
MulVAEK	80.69	61.38	76.17	73.57	77.20	54.40	65.50	54.01
GCN	78.25	56.50	70.06	59.35	79.84	59.68	66.67	47.13
GAT	71.88	43.76	70.88	61.11	80.03	60.06	61.60	45.42
RGCN	84.18	68.36	77.39	75.69	67.46	43.76	51.90	51.06
HAN	70.32	40.64	69.65	55.64	80.50	61.00	64.98	53.31
HGT	81.51	63.02	76.37	73.83	80.53	61.06	66.67	61.38
SeHGNN	72.00	44.00	66.80	51.14	82.47	64.94	71.31	65.12
GDAN	<u>84.49</u>	<u>68.98</u>	<u>78.82</u>	<u>76.79</u>	<u>84.07</u>	<u>68.14</u>	<u>70.04</u>	<u>65.39</u>
TrustVAE	**86.44**	**72.88**	**80.24**	**78.44**	**85.29**	**70.58**	**72.48**	**67.97**

optimal performance. Further increases in network depth did not lead to significant gains but introduced higher computational costs and greater complexity in hyperparameter tuning, consistent with observations reported in [26]. The latent space dimensionality was varied systematically between 8 and 256 to evaluate its influence on model effectiveness. For the ILRA module, the number of clusters was tested in the range of 2 to 6. In the BGAN component, the trust threshold was evaluated across values ranging from 0 to 1. The number of GNN layers was varied between 1 and 7 to examine its impact on performance. ReLU was selected as the activation function for all hidden layers owing to its superior performance compared with Tanh in our experiments. Given the binary classification nature of the SMEsD dataset, the final output layer used a sigmoid activation function. In contrast, softmax was applied in the multi-class setting in ECAD. Hyperparameter search ranges were defined as follows: the learning rate was explored within the logarithmic interval $[5 \times 10^{-4}, 1 \times 10^{-1}]$. Training was conducted for 400 epochs on SMEsD and 200 epochs on ECAD to ensure convergence, while avoiding overfitting.

5.2 Performance comparisons (RQ1)

As summarized in Table 2, TrustVAE achieves a robust performance in financial risk prediction and consistently outperforms existing SOTA methods. The key findings are discussed below.

- *Performance advantage of TrustVAE*. TrustVAE empirically outperformed all baseline models on both the SMEsD and ECAD datasets across multiple evaluation metrics. On the SMEsD dataset, TrustVAE improves over the second-best model by 1.95% in ROC-AUC, 3.90% in AR, 1.42% in Accuracy, and 1.65% in F1. On ECAD, it achieves improvements of 1.22% in ROC-AUC, 2.44% in AR, 1.17% in Accuracy, and 2.58% in F1. These consistent gains highlight TrustVAE's ability to extract informative structural risk features and capture the dynamics of risk propagation within supply chain networks. This capability is particularly important for risk-sensitive applications that require accurate ranking and classification. Its ROC-AUC scores of 86.44% on SMEsD and 85.29% on ECAD further confirm its strong discriminative power between high- and low-risk entities.

- *Comparison with other models.* Analyzing the performance across model categories provides deeper insights. Traditional Models (LogR, SVM, DT, and GBDT): These models exhibit limited predictive capability, particularly in handling graph-structured data and capturing nonlinear relationships. For instance, SVM achieves an F1 of only 10.53% on the SMEsD, highlighting its poor performance in supply chain scenarios. VAE-Based Variants (Ladder VAE, GraphVAE, GPGVAE, and VAEneu): Although some variants, such as MulVAEK, perform competitively on certain metrics (e.g., 75.57% F1 on SMEsD), their inconsistent results across datasets suggest limitations in stability and generalization. In contrast, TrustVAE integrates iterative clustering-based node aggregation, trust-aware edge propagation, and spatial-temporal graph autoregression within a unified architecture specifically designed for enterprise credit risk assessment in SCF scenarios. Graph-Based Models (GCN, GAT, and HAN): These models outperform traditional approaches but still fall short of TrustVAE. For example, HGT achieves an F1 of 73.83% on SMEsD, indicating that existing methods struggle to model heterogeneous graph structures.
- *Performance analysis across datasets.* To evaluate TrustVAE's adaptability, we assessed its performance on datasets with varying characteristics. Small-scale dataset (SMEsD): TrustVAE excels in ranking-based metrics, achieving 86.20% ROC-AUC and 72.40% AR. The 3.90% improvement in AR over GDAN underscores its superior capability for high-risk enterprise identification, a recall-critical task. Large-scale dataset (ECAD): Despite sparser node features, TrustVAE maintains a top-tier performance, particularly in AR (2.44% improvement over GDAN). Its F1of 67.97%, compared to GDAN's 65.39%, confirms its precision in low-information settings.

In summary, TrustVAE demonstrates both technological advancement and strong practical predictive capability, with superior performance across diverse datasets underscoring its value in real-world financial risk prediction.

5.3 Discussion of model variants (RQ2)

We conducted an ablation study to analyze the contribution of each TrustVAE module and the overall model efficiency. The variants were defined as follows:

- **TrustVAE (w/o ILRA)**: Excludes the ILRA module, relying only on iterative clustering for latent risk representation refinement.
- **TrustVAE (w/o BGAN)**: Omits the BGAN module, disabling bounded confidence message passing designed to suppress unreliable risk propagation.
- **TrustVAE (w/o TGAF)**: Removes the TGAF module, eliminating its capability to model dynamic credit risk flow across temporal and topological dimensions.
- **TrustVAE**: The complete model incorporating all modules.

As shown in Table 3, removing any of the three key modules leads to performance degradation, confirming each module's contribution to TrustVAE's effectiveness. The full model consistently outperformed all ablated variants, as detailed below for each component.

- *Impact of ILRA on performance*($Gain_{ILRA}$): Removing ILRA results in consistent performance drops across all evaluation metrics, most notably in AR, where an average decrease of 9.31% is observed across datasets. This substantial decline reflects the ILRA's critical role in refining latent risk representations through iterative learning and capturing cluster structures associated with different levels of credit risk. Additionally, the Accuracy drop is more evident on the ECAD dataset (3.10%) than on SMEsD (0.61%), indicating that ILRA enhances model robustness and discriminative capability in complex supply chain environments.

Table 3. The performance comparison of ablation study.

Dataset	Metric	w/o			TrustVAE	Gain		
		ILRA	BGAN	TGAF		ILRA	BGAN	TGAF
SMEsD	ROC-AUC	83.95	79.16	85.92	86.44	-2.49	-7.28	-0.52
	AR	67.90	58.32	71.84	72.88	-4.98	-14.56	-1.04
	Accuracy	79.63	77.60	75.15	80.24	-0.61	-2.64	-5.09
	F1	77.63	74.74	74.65	78.44	-0.81	-3.70	-3.79
ECAD	ROC-AUC	78.47	76.77	82.76	85.29	-6.82	-8.52	-2.53
	AR	56.94	53.54	65.52	70.58	-13.64	-17.04	-5.06
	Accuracy	69.38	61.63	71.71	72.48	-3.10	-10.85	-0.77
	F1	63.88	48.14	66.75	67.97	-4.09	-19.83	-1.22

- *Impact of BGAN on performance* ($Gain_{BGAN}$): BGAN demonstrates the most pronounced effect on prediction reliability, as reflected in the largest performance decreases in ROC-AUC for both SMEsD (-7.28%) and ECAD (-8.52%). These results are consistent with BGAN's core design objective of the BGAN, which is to mitigate unreliable risk propagation through bounded confidence message passing. Furthermore, the model's impact is more pronounced in large-scale credit networks, as evidenced by the greater degradation in AR observed on ECAD (-17.04%) than on SMEsD (-14.56%), indicating that BGAN plays an increasingly critical role in suppressing noisy or misleading risk signals in complex financial environments.
- *Impact of TGAF on performance* ($Gain_{TGAF}$): Although the absolute performance drops are relatively small, averaging 2.50% across metrics, removing TGAF reveals its essential role in modeling temporal dynamics. Notably, the 5.06% reduction in AR for ECAD contrasts sharply with the 1.04% drop in SMEsD, suggesting that TGAF is particularly effective at capturing time-dependent patterns in dynamic networks. Interestingly, a slight improvement of 0.52% in ROC-AUC was observed for SMEsD, which calls for further investigation into possible interactions between temporal and spatial features.

In summary, the performance gains confirm the essential role of each module: ILRA refines latent risk representations, BGAN curbs unreliable risk propagation through bounded confidence message passing, and TGAF captures dynamic credit risk flows across temporal and topological dimensions. The synergistic integration of these components underpins the superior performance of TrustVAE in supply chain finance risk prediction.

5.4 Discussion of in-depth studies (RQ3)

5.4.1 Flexibility of TrustVAE. To assess the generalizability of the TrustVAE's outputs, we integrated its predictions as enhanced features into various traditional models. As shown in Table 4, all the enhanced models consistently outperform their base counterparts across all metrics. This demonstrates that TrustVAE generates complementary high-value signals that can be effectively leveraged using conventional methods. Consequently, the integration framework enables the adoption of TrustVAE in classical machine learning pipelines, broadening its applicability to non-deep learning settings.

5.4.2 The analysis of ILRA module. In SCF, an enterprise's credit risk is influenced not only by its own operational performance but also by the complex transactional network and temporal evolution of relationships. A credit crisis in a core enterprise can rapidly propagate through upstream and downstream connections and trigger systemic risks. However, traditional approaches often rely on static financial data or simple statistical models, which fail to adequately capture the dynamic interactions among enterprises. More importantly, these methods typically treat enterprises as isolated entities, neglecting their positions, interaction patterns, and potential influence within the network, resulting in the omission of critical risk features. In contrast, our approach explicitly models

Table 4. A comprehensive evaluation of traditional methods, augmented by TrustVAE-generated insights, in tackling the financial risk prediction challenge

Model	SMEsD				ECAD			
	ROC-AUC	AR	Accuracy	F1	ROC-AUC	AR	Accuracy	F1 Score
LR	0.6726	0.3452	0.7699	0.6295	0.6767	0.3534	0.6034	0.4263
SVM	0.5221	0.0442	0.6538	0.1053	0.6565	0.3130	0.6076	0.4217
DT	0.7090	0.4180	0.7617	0.6113	0.6228	0.2456	0.5612	0.4856
GBDT	0.8300	0.6600	0.7413	0.4774	0.7784	0.5568	0.6160	0.5498
LR (GraphVAE)	0.7235	0.4470	0.7739	0.7351	0.6865	0.3730	0.6318	0.4465
SVM (GraphVAE)	0.5221	0.0442	0.6538	0.4453	0.6597	0.3194	0.6705	0.4740
DT (GraphVAE)	0.7307	0.4614	0.7719	0.7396	0.6657	0.3314	0.5814	0.5244
GBDT (GraphVAE)	0.8354	0.6708	0.7800	0.6351	0.7940	0.5880	0.6977	0.6031

the relational dependencies among enterprises through iterative learning, enabling the model to uncover the intrinsic data structure and discern natural clustering patterns. To further assess the model's ability to learn inherent structural properties, we analyze the evolution of inter-cluster distances during training. *Inter-class cosine similarity measurement.* The mean cosine similarity $\bar{S}_{\cos}$ is computed between all valid class pairs (C_i, C_j) $(0 \leq i < j < K$ where $|C_i||C_j| > 0)$ using robust weighted median representation vectors $\mathbf{v}_i = [\tilde{h}_d^{(i)}]_{d=1}^F$. Each dimension $\tilde{h}_d^{(i)}$ is derived through a weighted median computation:

$$\tilde{h}_d^{(i)} = \text{median}_{\mathbf{w}}\{h_d|\mathbf{h} \in C_i\}, \quad \mathbf{w} = \mathbf{1}_{|C_i|}, \tag{34}$$

where $\text{median}_{\mathbf{w}}$ denotes the weighted median operator with uniform weights $\mathbf{1}_{|C_i|}$ (a vector of ones). The final similarity metric is:

$$\bar{S}_{\cos} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left(\frac{\mathbf{v}_i^\top \mathbf{v}_j}{\|\mathbf{v}_i\|_2 \|\mathbf{v}_j\|_2} \right) \cdot \mathbb{I}[\|\mathbf{v}_i\|_2 > 0 \wedge \|\mathbf{v}_j\|_2 > 0]. \tag{35}$$

Advantages of ILRA in capturing inherent structure. To evaluate the ability of ILRA module to capture the inherent structure of financial data, we employed cluster analysis tools based on weighted median cosine similarity(Eq. 35). As shown in Fig. 3, experiments conducted on both the SMEsD and ECAD datasets (with 2-cluster and 3-cluster settings, respectively) reveal that the inter-class cosine similarity gradually decreases as the number of training epochs increases, exhibiting a stable and consistent downward trend. This reflects an increasing separation between clusters, demonstrating that the ILRA effectively enhances the distinguishability of the underlying data structures during training. These results demonstrate that the ILRA effectively captures and exploits intrinsic data structures, which inherently provides interpretability for credit risk assessment in SCF. Furthermore, this structural learning capability highlights its potential for modeling systemic risk propagation and heterogeneous enterprise interactions in complex financial ecosystems.

Comparative analysis of predictive performance. To evaluate the efficacy and robustness of the ILRA module, we integrated it into representative baseline models using two real-world datasets, as shown in Fig. 4. This experimental setup was designed to facilitate a comprehensive assessment of diverse data distributions and application scenarios. The selected models were chosen based on their representativeness across different modeling paradigms and their well-established performance in prior studies. Specifically, five prominent model architectures were selected for this study: GDAN [52], which leverages graph neural networks (GNNs) and has demonstrated strong performance in graph-based predictive tasks; and four VAE-based models: Vanilla VAE [43], Ladder VAE [43], Graph VAE [13], and GPGVAE [58]. The VAE-based models were included due

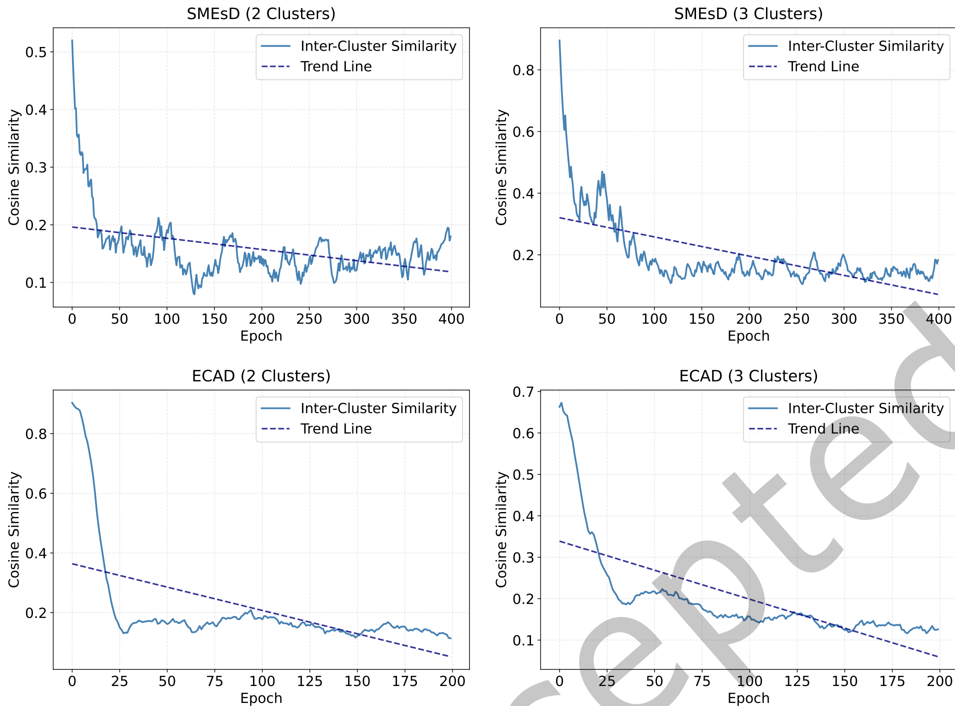


Fig. 3. Evolution of inter-class cosine similarity during ILRA training across datasets.

to their demonstrated ability to model complex data structures and uncertainty, as well as their widespread adoption across various domains. By incorporating both graph-based and VAE-based frameworks, we aim to provide a thorough evaluation of the adaptability and generalization capability of the ILRA module across various modeling paradigms. As illustrated in Fig. 4, the variants enhanced with ILRA (i.e., Base(ILRA)) consistently outperformed their original counterparts across all evaluation metrics, including ROC-AUC, AR, Accuracy, and F1. This consistent improvement demonstrates that integrating the ILRA module enhances the predictive performance of existing methods, supporting its flexibility and broad applicability across model architectures and datasets.

5.4.3 The analysis of BGAN module. Trust is a critical foundation for stable collaboration in SCF. A credit failure at a core enterprise can rapidly propagate through upstream and downstream linkages, potentially triggering systemic risks via cascading effects. However, the lack of mechanisms for quantifying trust at the network level hinders financial institutions' ability to capture its role in risk propagation, resulting in coarse approximations of the relational risk. Traditional risk propagation models typically ignore inter-enterprise trust as a key constraint or transmission weight, thereby yielding imprecise simulations of risk diffusion and a diminished sensitivity to risk stratification. In contrast, BGAN introduces a mechanism to quantify inter-enterprise trust dynamics during information exchange and incorporates them into credit risk forecasting in SCF. This enables the model to better capture how trust relationships influence credit risk propagation, offering a more comprehensive understanding of systemic risk in real-world business networks.

Advantages of BGAN for graph structure capturing. To investigate how different trust thresholds affect graph neural network modeling, this study constructs graphs based on enterprise trust relationships and applies edge filtering under various threshold settings. The Euclidean distance was used to quantify the trust levels between

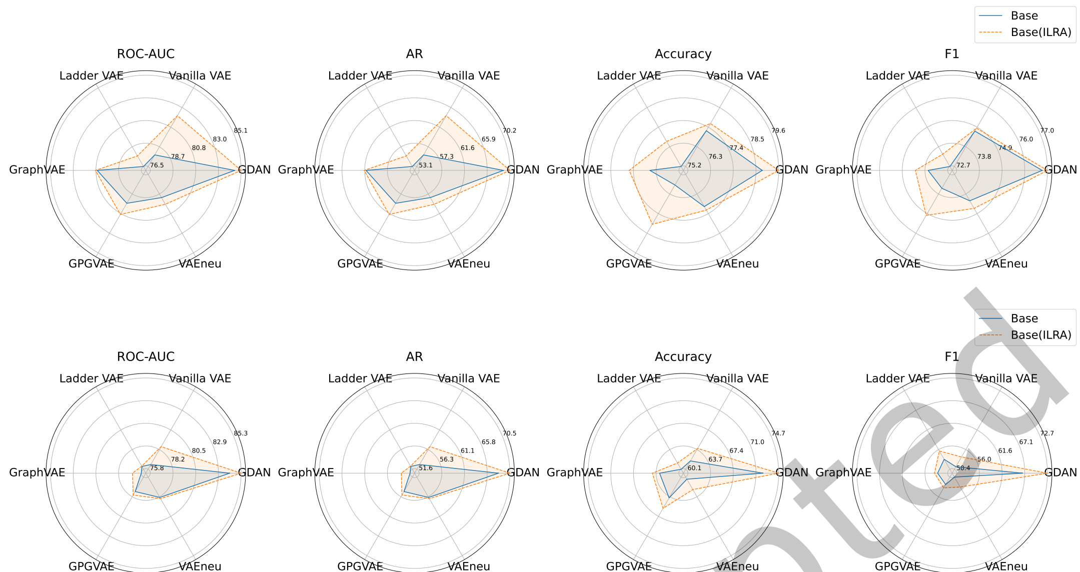


Fig. 4. Performance enhancement through ILRA integration across diverse models and datasets, with results depicted by blue and yellow lines (representing models without and with ILRA, respectively) and evaluated using a comprehensive set of metrics. The top plot corresponds to the SMEsD dataset, and the bottom one to the ECAD dataset.

nodes, with thresholds γ set at 0.1, 0.2, 0.3, 0.4, 0.5, 0.7, and 0.9. Only edges satisfying $d_{ij} \leq \gamma$ are retained, forming increasingly sparse subgraphs as the thresholds increase. The results demonstrate that higher trust thresholds lead to a substantial reduction in the number of edges and an overall sparser network structure, as shown in Fig. 5.

- *Number of connected components.* At lower thresholds, the graph contains numerous small disconnected components (497 when $\tau = 0.9$). As the threshold increases, these components merge, resulting in fewer but more tightly connected subgraphs (e.g., only 6 when $\gamma = 0.1$). This indicates that stricter trust constraints promote the formation of high-confidence and cohesive structures.
- *Average node degree.* Stricter thresholds reduce the number of edges retained per node, leading to a decline in the average node degree. For example, at $\gamma = 0.9$, enterprises maintain many connections, whereas at $\gamma = 0.1$, most have only a few trusted partners.

The results show that trust-based edge filtering significantly affects both global connectivity and local density in graph structures. Although this may limit global information flow, this approach allows models to focus on high-confidence subgraphs, improving the identification of key interaction patterns. This is especially valuable for tasks such as credit assessment and for anomaly detection. In summary, trust thresholds shape graph topology and, in turn, influence model performance in information aggregation and feature learning. Analyzing structural changes across thresholds provides deeper insights into the behaviors of graph-based models. These findings offer both theoretical support and practical implications for optimizing graph neural networks in real-world relational data applications.

Comparative analysis of node representation clustering performance. To intuitively evaluate the effectiveness of the learned enterprise node representations, we conducted a visualization analysis, which enhances model

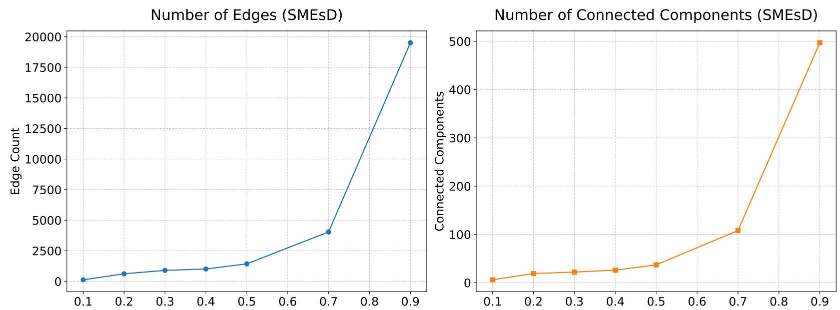


Fig. 5. Effect of trust thresholds on network sparsity and connectivity in the SMEsD dataset.

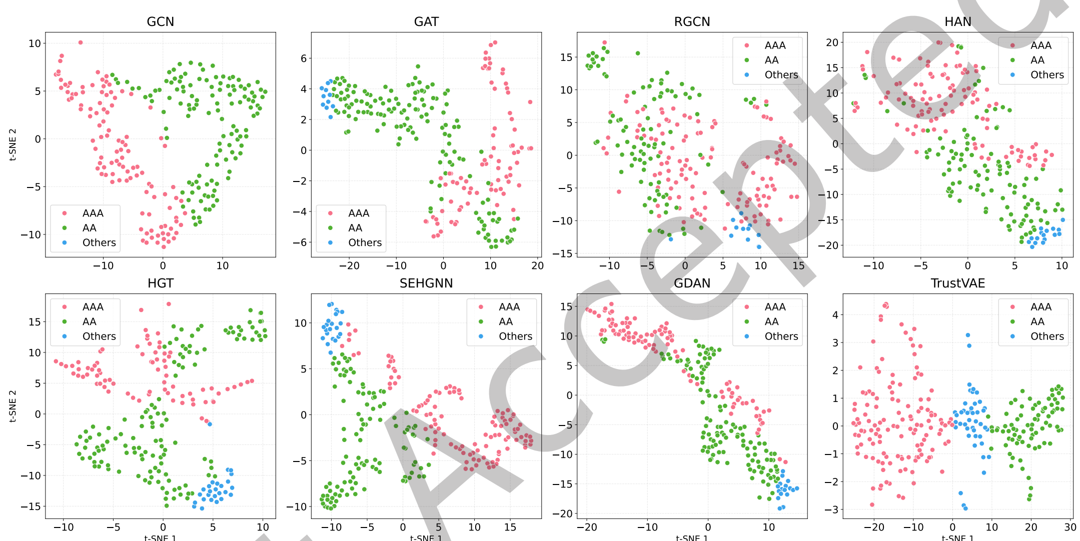


Fig. 6. TrustVAE outperforms GNNs in node representation clustering in the ECAD dataset.

interpretability through transparent structural patterns. Specifically, on the test set of the ECAD dataset, both the proposed TrustVAE model and several graph neural network baseline models were trained using the BGAN module to obtain trust-aware node representations of enterprises. Subsequently, we applied t-SNE to project these high-dimensional representations into a two-dimensional space for visualization. As shown in Fig. 6, TrustVAE produces highly clustered and well-separated representations of the three types of enterprises. Compared with the results from other models, the representations generated by TrustVAE exhibited significantly improved class separability, demonstrating its superior capability of learning effective and distinguishable node embeddings.

To further evaluate the clustering performance of TrustVAE, we applied the K-means algorithm to assign cluster labels to the learned node embeddings and then compared them with the ground-truth clusters. We adopted two widely used evaluation metrics: Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI). Given that K-means clustering is sensitive to initialization, we repeated the experiments ten times with different random seeds and report the average results in Table 5. The experimental results indicate that TrustVAE consistently outperforms all baseline methods in terms of clustering performance. In particular, compared to the

Table 5. Model performance (NMI/ARI) comparison

Metrics	GCN	GAT	RGCN	HAN	HGT	SEHGNN	GDAN	TrustVAE
NMI	0.0174	0.0377	0.0544	0.0137	0.0133	0.1366	0.1091	**0.2223**
ARI	-0.0005	0.0042	0.0759	-0.0019	-0.0006	0.0670	0.0221	**0.2738**

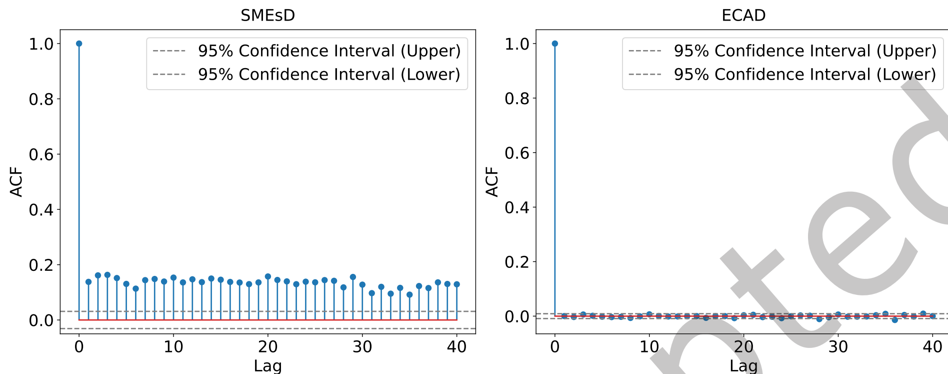


Fig. 7. ACF analysis of temporal dependencies in TGAF module across datasets.

best-performing baseline model SEHGNN, TrustVAE statistically improved both ARI and NMI scores, confirming its effectiveness in capturing meaningful structural patterns in enterprise graphs.

5.4.4 The analysis of TGAF module. Traditional machine learning algorithms face significant challenges in enterprise credit risk prediction due to the complexity and temporal dependency of financial data. Conventional methods, such as SVM and Random Forest, typically treat each instance independently, ignoring the temporal dependencies embedded in historical records (e.g., litigation and defaults). Consequently, these models fail to capture the relationship between current credit risk scores and past events, leading to suboptimal predictions over time. To address this limitation, we propose the TGAF module, which explicitly models the temporal dynamics to improve the forecasting accuracy. The experimental results support the following key findings:

Advantages of TGAF for temporal dependencies capturing. To evaluate the model's ability to capture temporal dependencies, we employed the autocorrelation function (ACF), which inherently enhances interpretability through transparent temporal pattern analysis. As illustrated in Fig. 7, the ACF values decay rapidly after lag 1, indicating short-term dependency and suggesting that an AR(1) model is suitable for capturing such patterns. In Fig. 7(left), the ACF values at lag 1 consistently exceed the 95% confidence interval, confirming the presence of strong temporal correlations in certain features. In contrast, Fig. 7(right) shows mixed results, where only some features exhibit significant temporal dependency, highlighting the model's capability to distinguish between time-sensitive and non-temporal features through interpretable autocorrelation patterns.

Comparative analysis of predictive performance. To evaluate the predictive capability of the TGAF module, we incorporated it into a set of representative VAE-based models and assessed their performance, as shown in Fig. 8. The baseline models include Vanilla VAE [43], Ladder VAE [43], Graph VAE [13], and GPGVAE [58], each of which was tested with and without TGAF enhancement. On the SMEsD dataset, all TGAF-augmented models achieved notable improvements over their baseline versions. For example, Vanilla VAE(TGAF) and GraphVAE(TGAF) attained ROC-AUC scores of 80.58% and 81.18%, along with AR scores of 61.16% and 62.36%, respectively. On

ECAD, although GPGVAE(TGAF) experienced a minor decline in the F1 score, it still achieved an ROC-AUC of 78.19% and an AR of 56.38%, further demonstrating the robustness of TGAF across diverse settings. These results collectively suggest that incorporating TGAF technology significantly optimizes the performance of VAE-based models. Nonetheless, practical applications require careful consideration of specific needs and the relative importance of individual performance metrics to ensure optimal model selection and deployment in practical applications.

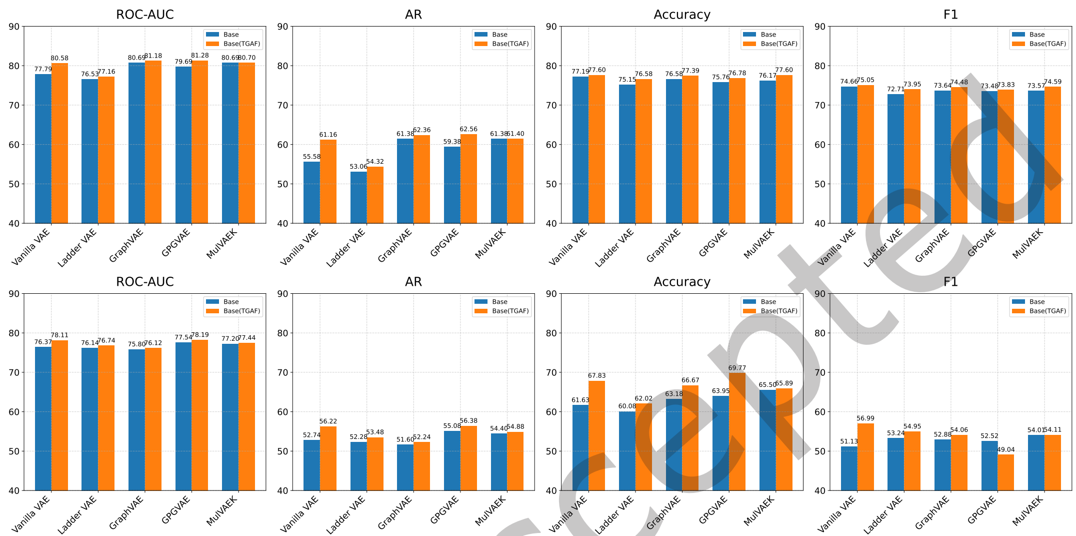


Fig. 8. Performance enhancement through TGAF integration across diverse model architectures and datasets, with results depicted by blue and yellow lines (representing models without and with TGAF, respectively) and evaluated using a comprehensive set of metrics. The top plot corresponds to the SMEsD dataset, and the bottom one to the ECAD dataset.

5.5 Hyperparameter analysis (RQ4)

To confirm the correlation between the model performance and specific experimental settings, we conducted additional experiments to investigate the sensitivity of the performance to various key hyperparameters.

Impact of the latent space size. To ascertain the influence of distinct latent space sizes on the functionality of financial risk prediction for various data groups, we evaluated the critical metrics that were appropriate for each set. The performance analysis of the SMEsD and ECAD frameworks revealed dataset-specific optimal latent dimensions, as shown in Fig. 9(a). SMEsD achieved peak metrics (Accuracy: 80.24%, F1: 78.76%, ROC-AUC: 85.97%, AR: 71.94%) at 128 dimensions, benefiting from the expanded representational capacity. ECAD showed divergent trends: while accuracy (70.93%) and F1-score (64.56%) peaked at 128 dimensions, ROC-AUC (83.23%) and AR (66.46%) maximized at 64 dimensions, reflecting metric-specific trade-offs. Both models degraded at extreme sizes (e.g., ECAD at 256 dimensions), indicating overfitting risks. This highlights the necessity of tailored latent space calibration—SMEsD thrives with moderately larger dimensions (128), while ECAD requires balanced granularity (64–128) to harmonize discriminative and generalizable features. Optimal sizing must align with data complexity and architectural objectives to enhance the predictive robustness.

Impact of the ILRA cluster number. In addition, to assess the influence of the number of experts on the prediction performance across diverse datasets, we scrutinized the pivotal metrics for each dataset, as shown in Fig. 9(b).

SMEsD peaked at 2 clusters (Accuracy: 80.24%, ROC-AUC: 86.21%), with performance declining sharply as clusters increased (e.g., 7.94% Accuracy drop at 3 clusters), indicating that oversegmentation harms simpler data. ECAD showed metric-dependent optima: accuracy/F1 peaked at 5 clusters (72.87%/67.32%), while ROC-AUC/AR maximized at 6 clusters (84.80%/69.60%), reflecting the need to balance granularity and stability. Both frameworks suffered instability at higher clusters (e.g., ECAD accuracy dropped 3.49% from 5 to 6), underscoring the risks of overclustering. The results emphasize tailored cluster selection: SMEsD favors minimal clusters (2–3) for homogeneity, while ECAD requires moderate granularity (5–6) to align metric priorities with heterogeneous data complexity.

Impact of the BGAN trust threshold. To explore how different trust thresholds influence the efficacy of the proposed approach on various datasets, we extensively analyzed the pertinent performance indicators, as illustrated in Fig. 9(c). SMEsD achieved balanced performance at 0.5–0.7 thresholds: Accuracy/F1 peaked at 0.5 (77.80%/75.01%), while ROC-AUC/AR maximized at 0.7 (84.22%/68.44%), indicating moderate thresholds enhance reliability without over-filtering. Higher thresholds (e.g., 0.9) caused accuracy/F1 to decline, suggesting that excessive strictness harms decision cohesion. ECAD exhibited conflicting trends: accuracy/F1 peaked at 0.9 threshold (74.42%/68.93%), yet ROC-AUC/AR were highest at 0.1 threshold (85.29%/70.58%), reflecting a trade-off between precision and robustness. Intermediate thresholds (0.3–0.7) degraded all metrics, implying sensitivity to noise suppression. The results emphasize dataset-specific calibration: SMEsD benefits from moderate thresholds (0.5–0.7) to balance discriminative power, while ECAD requires threshold prioritization (e.g., 0.1 for robustness vs. 0.9 for accuracy) based on task-critical metrics. Overly strict thresholds risk discarding informative patterns, necessitating context-aware optimizations.

Impact of the BGAN layer number. To evaluate the sensitivity of the proposed method to layer number variations across datasets, we systematically measured its efficacy through critical performance metrics, as visualized in Fig. 9(d). SMEsD achieved peak accuracy (80.24%)/F1 (79.09%) at 2 layers and highest ROC-AUC (86.44%)/AR (72.88%) at 5 layers, suggesting that moderate depth balances feature learning. However, 7 layers caused sharp Accuracy decline (6.11% drop), indicating overfitting risks. ECAD peaked uniformly at 2 layers (accuracy:72.48%, ROC-AUC:85.29%), with deeper layers degrading performance (e.g., 67.44% accuracy at 3 layers). A partial recovery at 7 layers (ROC-AUC:84.40%) hinted at limited adaptability to complexity. The results emphasize dataset-driven design: SMEsD tolerates moderate depth (2–5 layers) for nuanced patterns, while ECAD thrives with shallow architectures (2 layers) to avoid instability. Excessive layers harm generalization, necessitating architecture-data alignment.

Table 2 and the parameter sensitivity analyses show that TrustVAE consistently outperforms the baselines, even with suboptimal hyperparameters. This indicates that the TrustVAE maintains high performance stability across a wide range of settings. Therefore, TrustVAE can be considered a robust and reliable solution for a specific task.

6 Conclusion and Future Work

This study addresses a critical gap in existing credit risk prediction frameworks for SCF, which commonly overlook the systemic interconnectedness and social embeddedness inherent in enterprise networks. To overcome this limitation, we propose TrustVAE, a novel generative framework designed for SMEs credit risk assessment. Specifically, the ILRA module learns structured and interpretable risk representations via iterative clustering and nonlinear transformations, and extracts latent risk patterns from heterogeneous enterprise data. In parallel, BGAN introduces a trust-aware message passing mechanism based on bounded confidence, dynamically modulating risk diffusion using trust derived from transactional and event data, thus enhancing the robustness and interpretability of relational modeling in GNNs. Furthermore, the TGAF module captures joint spatio-temporal dynamics through a flow-based autoregressive architecture, enabling multi-scale modeling of risk evolution over time. To deepen

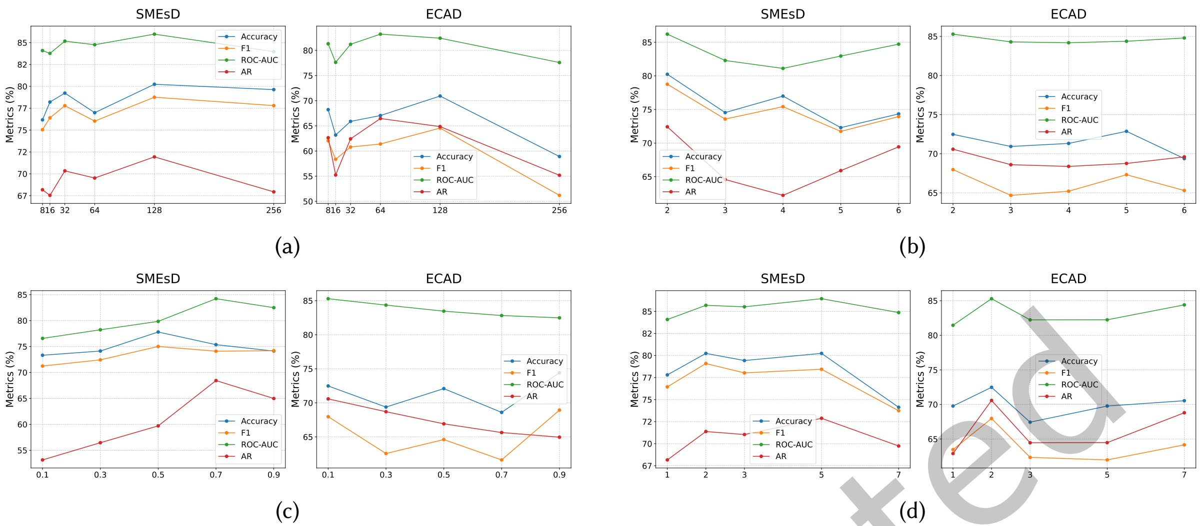


Fig. 9. Parameter sensitivity analysis across two datasets: (a) Latent space size effect, (b) ILRA cluster number effect, (c) BGAN trust threshold impact, (d) BGAN layer number influence.

our understanding of the contributions of various modules and assess their suitability under real-world data conditions, we conducted a series of meticulous studies, including ablation analyses, in-depth investigations, efficiency evaluations and core module migration tests. These evaluations highlight TrustVAE's applicability in real-world data conditions. Future work will focus on enhancing the model's scalability, generative diversity, security and interpretability through the integration of non-financial signals and adaptive graph learning to broaden its applicability in large-scale financial environments.

References

[1] Afshin Ashofteh and Jorge M Bravo. 2021. A conservative approach for online credit scoring. *Expert Systems with Applications* 176 (2021), 114835.

[2] Allen N Berger and Gregory F Udell. 1995. Relationship lending and lines of credit in small firm finance. *Journal of Business* (1995), 351–381.

[3] Kejun Bi, Chuanjie Liu, and Bing Guo. 2025. Enterprise risk assessment model based on graph attention networks. *Applied Intelligence* 55, 3 (2025), 1–17.

[4] Txus Blasco, J Salvador Sánchez, and Vicente García. 2024. A survey on uncertainty quantification in deep learning for financial time series prediction. *Neurocomputing* 576 (2024), 127339.

[5] Yukuo Cen, Jianwei Zhang, Xu Zou, Chang Zhou, Hongxia Yang, and Jie Tang. 2020. Controllable multi-interest framework for recommendation. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2942–2951.

[6] Stefano Cenni, Stefano Monferrà, Valentina Salotti, Marco Sangiorgi, and Giuseppe Torluccio. 2015. Credit rationing and relationship lending. Does firm size matter? *Journal of Banking & Finance* 53 (2015), 249–265.

[7] Gourab Chakraborty. 2020. Evolving profiles of financial risk management in the era of digitization: The tomorrow that began in the past. *Journal of Public Affairs* (2020).

[8] Yu Chen, Lingfei Wu, and Mohammed Zaki. 2020. Iterative deep graph learning for graph neural networks: Better and robust node embeddings. *Advances in neural information processing systems* 33 (2020), 19314–19326.

[9] Elena Dumitrescu, Sullivan Hué, Christophe Hurlin, and Sessi Tokpavi. 2022. Machine learning for credit scoring: Improving logistic regression with non-linear decision-tree effects. *European Journal of Operational Research* 297, 3 (2022), 1178–1192.

[10] Bo-Jing Feng, Xi Cheng, Hao-Nan Xu, and Wen-Fang Xue. 2024. Corporate credit ratings based on hierarchical heterogeneous graph neural networks. *Machine Intelligence Research* 21, 2 (2024), 257–271.

[11] Sergio Genovesi, Julia Maria Mönig, Anna Schmitz, Maximilian Poretschkin, Maram Akila, Manoj Kahdan, Romina Kleiner, Lena Krieger, and Alexander Zimmermann. 2024. Standardizing fairness-evaluation procedures: interdisciplinary insights on machine learning algorithms in creditworthiness assessments for small personal loans. *AI and Ethics* 4, 2 (2024), 537–553.

[12] Mark Granovetter. 1985. Economic action and social structure: The problem of embeddedness. *Amer. J. Sociology* 91, 3 (1985), 481–510.

[13] Jiawei He, Yu Gong, Joseph Marino, Greg Mori, and Andreas Lehrmann. 2018. Variational autoencoders with jointly optimized latent dependency structure. In *International Conference on Learning Representations*.

[14] Michael G Hertzel, Zhi Li, Micah S Officer, and Kimberly J Rodgers. 2008. Inter-firm linkages and the wealth effects of financial distress along the supply chain. *Journal of Financial Economics* 87, 2 (2008), 374–387.

[15] Ziniu Hu, Yuxiao Dong, Kuansan Wang, and Yizhou Sun. 2020. Heterogeneous graph transformer. In *The World Wide Web Conference*. 2704–2710. doi:10.1145/3366423.3380027

[16] Guanming Huang, Zhen Xu, Zhenghao Lin, Xiaojun Guo, and Mohan Jiang. 2024. Artificial Intelligence-Driven Risk Assessment and Control in Financial Derivatives: Exploring Deep Learning and Ensemble Models. *Transactions on Computational and Scientific Methods* 4, 12 (2024).

[17] Yating Huang, Zhao Wang, and Cuiqing Jiang. 2024. Diagnosis with incomplete multi-view data: A variational deep financial distress prediction method. *Technological Forecasting and Social Change* 201 (2024), 123269.

[18] Cuixia Jiang, Haijing Gao, and Qifa Xu. 2024. China's risk contagion using the mixed-frequency macro-financial network. *Economic Systems* 48, 4 (2024), 101212.

[19] Wenjun Jiang, Guojun Wang, Md Zakirul Alam Bhuiyan, and Jie Wu. 2016. Understanding graph-based trust evaluation in online social networks: Methodologies and challenges. *Acm Computing Surveys (Csur)* 49, 1 (2016), 1–35.

[20] Thomas N. Kipf and Max Welling. 2017. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*.

[21] Alireza Koochali, Ensiye Tahaei, Andreas Dengel, and Sheraz Ahmed. 2025. VAEneu: A new avenue for VAE application on probabilistic forecasting. *Applied Intelligence* 55, 6 (2025), 1–23.

[22] Stefan Lessmann, Bart Baesens, Hsin-Vonn Seow, and Lyn C Thomas. 2015. Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research* 247, 1 (2015), 124–136.

[23] Xiaoyang Li, Weimin Li, Xiao Yu, Zhongming Han, and Qun Jin. 2025. Financial risk assessment of imbalanced data based on nonlinear causal time-series network. *Information Processing & Management* 62, 3 (2025), 104025.

[24] Yijing Li, Jing Tang, Sinan Gonul, and Danielle Barbe. 2025. A novel approach for capturing consumer behavior and preference-MulVAEK. *IEEE Computational Intelligence Magazine* 20, 2 (2025), 33–44.

[25] Youru Li, Zhenfeng Zhu, Xiaobo Guo, Shaoshuai Li, Yuchen Yang, and Yao Zhao. 2023. HGV4Risk: Hierarchical global view-guided sequence representation learning for risk prediction. *ACM Transactions on Knowledge Discovery from Data* 18, 1 (2023), 1–21.

[26] Dawen Liang, Rahul G Krishnan, Matthew D Hoffman, and Tony Jebara. 2018. Variational autoencoders for collaborative filtering. In *Proceedings of the 2018 World Wide Web Conference*. 689–698. doi:10.1145/3178876.3186150

[27] Yancheng Liang, Jiajie Zhang, Hui Li, Xiaochen Liu, Yi Hu, Yong Wu, Jinyao Zhang, Yongyan Liu, and Yi Wu. 2023. DeRisk: An effective deep learning framework for credit risk prediction over real-world financial data. In *Proceedings of the Fifth Workshop on Financial Technology and Natural Language Processing and the Second Multimodal AI For Financial Forecasting*. 81–93.

[28] Marco Locurcio, Francesco Tajani, Pierluigi Morano, Debora Anelli, and Benedetto Manganelli. 2021. Credit risk management of property investments through multi-criteria indicators. *Risks* 9, 6 (2021), 106.

[29] Jianxin Ma, Chang Zhou, Peng Cui, Hongxia Yang, and Wenwu Zhu. 2019. Learning disentangled representations for recommendation. *Advances in Neural Information Processing Systems* 32 (2019).

[30] Rogelio A Mancisidor, Michael Kampffmeyer, Kjersti Aas, and Robert Jenssen. 2022. Generating customer's credit behavior with deep generative models. *Knowledge-Based Systems* 245 (2022), 108568.

[31] Joseph Marino, Lei Chen, Jiawei He, and Stephan Mandt. 2020. Improving sequential latent variable models with autoregressive flows. In *Symposium on Advances in Approximate Bayesian Inference*. PMLR, 1–16.

[32] Andrea Moro and Matthias Fink. 2013. Loan managers' trust and credit access for SMEs. *Journal of banking & finance* 37, 3 (2013), 927–936.

[33] Kavitha Muthukumaran, K Hariharanath, and Vani Haridasan. 2023. Feature selection with optimal variational auto encoder for financial crisis prediction. *Computer Systems Science & Engineering* 45, 1 (2023).

[34] Patrick Asante Owusu, Etienne Tajeuna, Jean-Marc Patenaude, Armelle Brun, and Shengrui Wang. 2023. Rethinking temporal dependencies in multiple time series: A use case in financial data. In *2023 IEEE International Conference on Data Mining (ICDM)*. IEEE, 1247–1252. doi:10.1109/ICDM58522.2023.00156

[35] Sulin Pang, Xianyan Hou, and Lianhu Xia. 2021. Borrowers' credit quality scoring model and applications, with default discriminant analysis based on the extreme learning machine. *Technological Forecasting and Social Change* 165 (2021), 120462.

[36] Mitchell A Petersen and Raghuram G Rajan. 1994. The benefits of lending relationships: Evidence from small business data. *The Journal of Finance* 49, 1 (1994), 3–37.

[37] Jing Quan and Xuelian Sun. 2024. Credit risk assessment using the factorization machine model with feature interactions. *Humanities and Social Sciences Communications* 11, 1 (2024), 1–10.

[38] Antti Rasmus, Mathias Berglund, Mikko Honkala, Harri Valpola, and Tapani Raiko. 2015. Semi-supervised learning with ladder networks. *Advances in Neural Information Processing Systems* (2015).

[39] Michael Schlichtkrull, Thomas N Kipf, Peter Bloem, Rianne Van Den Berg, Ivan Titov, and Max Welling. 2018. Modeling relational data with graph convolutional networks. In *European Semantic Web Conference*. Springer, 593–607.

[40] Qigan Shao, Changchang Jiang, James JH Liou, Peiyao Su, Ying Yuan, and Zhu Dan. 2025. Exploring the influencing factors of digital transformation: empirical results from SMEs in China. *Managerial and Decision Economics* 46, 4 (2025), 2364–2380.

[41] Yong Shi, Yi Qu, Zhensong Chen, Yunlong Mi, and Yunong Wang. 2024. Improved credit risk prediction based on an integrated graph representation learning approach with graph transformation. *European Journal of Operational Research* 315, 2 (2024), 786–801.

[42] Naeem Siddiqi. 2012. *Credit risk scorecards: developing and implementing intelligent credit scoring*. Vol. 3. John Wiley & Sons.

[43] Casper Kaae Sønderby, Tapani Raiko, Lars Maaløe, Søren Kaae Sønderby, and Ole Winther. 2016. Ladder variational autoencoders. *Advances in Neural Information Processing Systems* 29 (2016).

[44] Fei Tan, Xiurui Hou, Jie Zhang, Zhi Wei, and Zhenyu Yan. 2018. A deep learning approach to competing risks representation in peer-to-peer lending. *IEEE Transactions on Neural Networks and Learning Systems* 30, 5 (2018), 1565–1574.

[45] Yandan Tan, Hongbin Zhu, Jie Wu, and Hongfeng Chai. 2024. DPTVAE: Data-driven prior-based tabular variational autoencoder for credit data synthesizing. *Expert Systems with Applications* 241 (2024), 122071.

[46] Hirofumi Uchida, Gregory F Udell, and Nobuyoshi Yamori. 2012. Loan officers and relationship lending to SMEs. *Journal of Financial Intermediation* 21, 1 (2012), 97–122.

[47] Dejan Varmedja, Mirjana Karanovic, Srdjan Sladojevic, Marko Arsenovic, and Andras Anderla. 2019. Credit card fraud detection-machine learning methods. In *2019 18th International Symposium INFOTEH-JAHORINA (INFOTEH)*. IEEE, 1–5.

[48] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph attention networks. In *International Conference on Learning Representations*.

[49] Jianfei Wang, Cuiqing Jiang, Lina Zhou, and Zhao Wang. 2024. Assessing financial distress of SMEs through event propagation: An adaptive interpretable graph contrastive learning model. *Decision Support Systems* 180 (2024), 114195.

[50] Jiaxing Wang, Guoquan Liu, Xiaobo Xu, and Xinjie Xing. 2024. Credit risk prediction for small and medium enterprises utilizing adjacent enterprise data and a relational graph attention network. *Journal of Management Science and Engineering* 9, 2 (2024), 177–192.

[51] Xiao Wang, Houye Ji, Chuan Shi, Bai Wang, Yanfang Ye, Peng Cui, and Philip S Yu. 2019. Heterogeneous graph attention network. In *The World Wide Web Conference*. 2022–2032. doi:10.1145/3308558.3313562

[52] Shaopeng Wei, Beni Egressy, Xingyan Chen, Yu Zhao, Fuzhen Zhuang, Roger Wattenhofer, and Gang Kou. 2024. Graph dimension attention networks for enterprise credit assessment. *arXiv preprint arXiv:2407.11615* (2024).

[53] Shaopeng Wei, Jia Lv, Yu Guo, Qing Yang, Xingyan Chen, Yu Zhao, Qing Li, Fuzhen Zhuang, and Gang Kou. 2024. Combining intra-risk and contagion risk for enterprise bankruptcy prediction using graph neural networks. *Information Sciences* 659 (2024), 120081.

[54] Jin Xiao, Yu Zhong, Yanlin Jia, Yadong Wang, Ruoyi Li, Xiaoyi Jiang, and Shouyang Wang. 2024. A novel deep ensemble model for imbalanced credit scoring in internet finance. *International Journal of Forecasting* 40, 1 (2024), 348–372.

[55] Xiaofeng Xie, Fengying Zhang, Li Liu, Yang Yang, and Xiuying Hu. 2023. Assessment of associated credit risk in the supply chain based on trade credit risk contagion. *PloS One* 18, 2 (2023), e0281616.

[56] Tianyi Yang, Ang Li, Jiahao Xu, Guangze Su, and Jiufan Wang. 2024. Deep learning model-driven financial risk prediction and analysis. *Applied and Computational Engineering* 77 (2024), 196–202.

[57] Xiaocheng Yang, Mingyu Yan, Shirui Pan, Xiaochun Ye, and Dongrui Fan. 2023. Simple and efficient heterogeneous graph neural network. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 10816–10824. doi:10.1609/AAAI.V37I9.26283

[58] Xue Yu, Muchen Li, Yan Leng, and Renjie Liao. 2024. Learning latent structures in network games via data-dependent gated-prior graph variational autoencoders. In *Forty-first International Conference on Machine Learning*, Vol. 235. 57507–57526. doi:10.5555/3692070.3694442

[59] Wen Zhang, Shaoshan Yan, Jian Li, Xin Tian, and Taketoshi Yoshida. 2022. Credit risk prediction of SMEs in supply chain finance by fusing demographic and behavioral data. *Transportation Research Part E: Logistics and Transportation Review* 158 (2022), 102611.

[60] Xiaoming Zhang and Lean Yu. 2024. Consumer credit risk assessment: A review from the state-of-the-art classification algorithms, data traits, and learning methods. *Expert Systems with Applications* 237 (2024), 121484.

[61] Jianan Zhao, Xiao Wang, Chuan Shi, Binbin Hu, Guojie Song, and Yanfang Ye. 2021. Heterogeneous graph structure learning for graph neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 4697–4705.

[62] Yang Zhao, John W Goodell, Yong Wang, and Mohammad Zoynul Abedin. 2023. Fintech, macroprudential policies and bank risk: Evidence from China. *International Review of Financial Analysis* 87 (2023), 102648.

[63] Yanqiao Zhu, Weizhi Xu, Jinghao Zhang, Yuanqi Du, Jieyu Zhang, Qiang Liu, Carl Yang, and Shu Wu. 2021. A survey on graph structure learning: Progress and opportunities. *arXiv preprint arXiv:2103.03036* (2021).